Review dossier S: the sciences

Reader edition of the work at commit f0c8f7e69, 10 November 2026. Links into the repository are plain text here; the covering letter says how to reach it.

The theory is a map of how anything forms, persists, changes and ends, drawn first for alignment: what a selection reaches, what it drains beyond what it reads, how it locks on what it already chose. This dossier reads frameworks of the sciences onto it, where they meet it and where they do not. It is for physicists working on thermodynamics, evolutionary biologists and population geneticists, ecologists and resource economists, computational neuroscientists, control theorists and researchers of complex systems. No independent human review of this work is recorded yet. It was written with AI, the Lean proofs included, and the person who directs the work has read its important parts, not every word. The proofs and quotations are checked by machine, everything else by further AI readers (how this dossier was made). It assumes the theory: for what the theory is as a whole, read the one page first.

The dossier in ten minutes

The question. The theory starts from relations and the amounts exchanged along them, and asks how patterns form, last, change and end. It claims to *contain* frameworks of the sciences. A containment has three parts: the framework's premises, set out on the theory's premise card; a theorem in Lean stating the framework's law as an instance; and something the theory forbids that the framework leaves open. Without the third part a containment is a restatement. Does a containment here state your framework correctly, and does what it forbids go beyond what your field already forbids?

What it does here, and the verdict on prior art (S-01, S-02). It reads conservation, selection limited by observation and the comparison of opportunity sets in one vocabulary; by an AI review against the literature, none of its central results is new mathematics, each being near-definitional or a known result restated, and the premise ledger is new only as an artefact. The theory reaches farther than the few contained frameworks that lack conservation, renewal over time or a threshold, by joining them to its own premises and laws, though an AI audit found that each statement so forbidden is known in a neighbouring field. Than the free-energy principle alone it reaches farther in what it forbids, five statements, two of them weakly, each already known elsewhere, and it states less where it states nothing yet, in densities, continuous time and deep models among others. With the rest it is level or states less, and it does not yet reach farther than the sciences taken together. The literature verdicts come from the AI's prior knowledge and earlier prior-art checks, and an AI literature search in three rounds found no result that no neighbouring field states: a bounded search, neither established novelty nor established absence.

What a law here is (08 §10). Most laws are theorems about models in whole numbers and discrete steps, under hypotheses each item names; the periphery's are over Mathlib's reals, their axioms reported. They are the theory's laws, not yet any science's: one with content can become a science's law for a domain only once the domain is shown to meet its hypotheses, through a stated correspondence that reads the model's terms in measured quantities and says what its modelling choices leave out. The root's laws hold of every relation, so they forbid none; a law with content forbids through its hypotheses.

What would refute what. No measurement refutes a theorem: it falls by a failed Lean check, or by a statement that does not say what its item says. A law's conclusion failing in a domain shows that the domain does not meet its hypotheses or that the correspondence misreads them, not that the theorem is

false [AI reading]. The theses exclude observations, each frozen test scored by a rule fixed before the data (S-03).

Thesis	How far seen	Forbids (its item has the rest)	Its frozen tests so far
T1: persistence certifies the joint draw	in the model proved: on the regime, in whole numbers, under the premises S-12's block names, a level up to a top `hi` the step keeps ends exactly where it starts below `H + y`, `y` a level below which giving falls short of the draw, and the point of no return, where `H + y` is at most `hi`, is `H + y`, and, the draw positive and met at `y`, the point lies at least the draw above a need where giving stops below it, and on the World, under the gate at need `m` and the premises S-11 names, `n` consecutive steps of joint excess from a level below `n + m` end every formation depending on the condition at a need of at least `m`, whatever each drawer's own draw; beyond the model conjectured	a condition below a need measured in advance returning with nothing given from beyond; a dependent active while its condition stood below its need	of the three settings passed, one of real data (regimes of one yeast experiment held out from the fit, not from the analyst, who had seen their fates separate post hoc; the dilution the only draw) and two simulations, the only tests whose design has more than one drawer (in the simulated yeast the passes bearing on T1 test the measured quantities, save the two harvesters' super-additive harms; in the commons joint excess is an identity of its accounting, and the scope prediction, with one harvester, held); on yeast values not read before no frozen test of the need or its fates has passed; failed in four, among them fish stocks under held catches and a salt shock, after which, on the first reading, fates separated no better than by a null, the letter of T1's *Refuted by*, every population standing above the need, so that what T1 forbids was not reached; no field test of a need measured in advance: two designed on Wisconsin's walleye lakes each ran once and refused before reading a row, saying nothing of T1

Thesis	How far seen	Forbids (its item has the rest)	Its frozen tests so far
T2: the state of a renewable condition under a draw is its level, its recent giving and one lag, its place named before the data	conjectured; untested on a wild condition outside fisheries; its former wording failed in the flour beetle's adult-loss cultures; narrowed on 6 November 2026 on that failure, which forbids less	a fourth variable from the level's own history removing a tenth of the residual variance in the median series; the level alone forecasting as well as the three numbers at five steps of the grain they are read at, fixed before the data, in a series not shown by a bound to sit at the regime's-step end; the lag's coefficients changing sign between series	of four frozen fish-stock tests, one's verdicts on the state and the lag passed (its two on the draw's response failed), one failed its state verdict and one missed its bound; in the fourth (UE32) T2's numbers beat the field's fitted embedding by its frozen criterion, both behind persistence in most stocks, what passed being sufficiency against it at these lengths, not the lag's place or T2's law; in the flour beetle, where a stage before recruitment is counted, its former wording "as far as anything measured does" failed on a frozen run (UE31b, run on 6 November 2026; UE31's own run, on 3 November, decided nothing, its adult-loss cultures too short for its rules) to the pupae and the adults, a class of fewer variables (08 §9's defeater, in that domain only), leaving "as far as the level's own history does", with the lag two censuses back, where the life cycle puts it: there the three numbers tie the field's state on the adults' course by construction, as that field's model predicts, not T2 passed; the lag's sign, their one contrary prediction, was not decided (S-15)
T6: a break is seen coming only through what couples into what breaks	the invisible half proved on the World model; the seen half proved in the periphery, in the linearized and logistic models; one law across domains, and "only where" beyond these models, conjectured	given the stock's own course, its early-warning statistics separating runs in which a dependent's need is crossed from runs in which it is not, the dependent taking no measurable share	three on seabirds, two sharing rows, did not reject it at their thresholds, one at low power with estimates in the refuting direction; none establishes conditional independence

T3's law for optima is proved (S-13, S-20), across domains conjectured; in simulation, before T3 was worded, the draw against margin forecast an exact best response's side effects (8 of 8) and failed for tabular learners (80 of 151; MQ1); its clause on trained systems, conjectured with partial support, is the alignment dossier's.

What is not claimed. A science's law. That actual systems meet the models' premises, which is not shown. That the root's calculus explains how things go on, which is argued, not proved, and on a first

count a rival with no word for continuation ties it (08 §9). The account's ethics, which emerges from continuation, is not treated here but in the philosophy dossier; where a result ranks, its theorem states the hypotheses it takes, such as the continuation condition (for anything, its continuation into the future has significance).

Where to read, by field. Physics, S-04 to S-06. Evolution, S-07 to S-10. Ecology and renewable resources, S-11 to S-16. Mind and inference, S-17 to S-20, with the free-energy principle in the theory. Order and open-endedness, S-21 to S-23.

What we ask, and how long. A verdict on each item in the response form: whether the containment is correctly stated, its premises set out and the Lean saying what the item says; whether it is credited where it restates known work and put at its right strength; whether its frozen tests discriminate; and what is missing. Any item can be answered alone. This section takes thirteen to seventeen minutes. With an hour, read your field's items, and S-02 unless they are S-11 to S-16, which with this section take 65 to 85 minutes, 55 to 75 if S-16 is left for later. The whole takes 176 to 230 minutes; a reply on every item, a day or more.

Reading the items

Conventions. Lean paths are relative to `experiments/integrated-correction/general-framework/generative-basis/lean/becoming/` for the pure core, and periphery files to `experiments/integrated-correction/general-framework/generative-basis/lean/`. Lean blocks give statements exactly, proofs cut at `:=`, definitions whole. They are given for the central items only; the rest are named. No Lean is needed to review: each item's **Meaning** says in words what is stated. The pure core compiles from Lean's prelude alone, and every source declaration in it has an empty axiom report: "An empty report shows that no logical axiom is used. It says nothing about whether the modelling fits an actual unfolding" (ch. 8). The modelling choices are listed as commitments of the representation (08 §4), ten in all, among them discrete steps on one shared index, natural-number amounts (a positive amount is at least one unit), activity as a proposition, requirements as thresholds, the naming of relata, and identity by naming (which formation continues which is named), on which S-21 and S-22 turn, their main laws being read by name. Many laws below depend on natural-number amounts and say so.

The terms:

- **Formation, condition, need, depends.** A *formation* is a pattern that goes on: an organism, a lineage, a population, a fishery, a firm. A *condition* is an amount it draws on, such as a stock, a nutrient or available work. A formation *depends* on a condition at a *need* when it is active only while the condition stands at or above that need. The dependence is *broken* when the condition falls below the need.
- **Level, draw, giving, renewal.** A condition's *level* is how much of it stands at a step, a whole number. A *draw* is what is taken from it across a step, *giving* what is given to it, and *renewal* giving read as a law of the level.
- **The World and its premises.** The *World* is the model of relata exchanging amounts across discrete steps. Its premises are properties a world may have or lack, among them *balance* (every change of a level is what is taken from it and given to it), *dissipation* (on every family without repetition closed across a step, a formation that takes gives back at least one unit less), *activity needing support*, *only active relata give*, *embodiment* and a *census* of the active relata (the rest theorem also needs a census of what

stands, S-06). A family is *closed* across a step when no formation taking part exchanges with a condition outside it; a step is *enclosed around* a family when some finite family without repetition containing it is closed.

- **The theory's first law.** Across every step no formation gives a closed family without repetition more than it takes from it (`Flow.Gainless`). Under the exergy balance, thermodynamics' second law read in available work gives it (`EntropyProduction.gainless_of_secondLaw`; no converse is proved); it is not thermodynamics' first law, which S-04's balance theorem reads.
- **The gate, joint excess.** A renewal is *gated* at `m` when every formation giving to the condition depends on it at a need of at least `m`, so the renewal depends on what it renews and nothing renews the condition below `m`. A step of *joint excess* is one at which all takers together take more than the step gives.
- **Regime, point of no return.** A *regime* is one condition under a standing draw `H`, its next level `(x − H) + g(x − H)`: the draw taken first, as much as there is, then giving by a law `g` of what is left. Its *point of no return* is the least level from which it does not end. A *margin* is how far a level lies above a need.
- **Couples into, room, rising, optimum.** A quantity *couples into* a selector when the selector's signal depends on it; what does not couple in lies *beyond*. An order on outcomes is *rising* when it ranks above every outcome exceeding another on what couples in, whatever either leaves beyond. An *optimum* is an eligible outcome no eligible outcome ranks above. *Room* holds when every eligible outcome leaving more beyond is beaten, on what couples in, by an eligible outcome.
- **Widening.** One course *widens* another when at every step it admits every formation the other admits, and at some step one more.
- **Restated, near-definitional.** A result is *restated* when it is a known result in the theory's terms, *near-definitional* when it follows almost at once from its definitions.
- **Labels.** *(root)*: proved on the theory's grammar, for every carrier, relation and class. *(any type)*, *(any process)*: proved for every type or step relation. *(World)*: proved on the World model, with its premises and local hypotheses named. *(the regime)*, *(any `R`)*: proved for one condition under any giving law, or for any renewal law `R`, in whole numbers. *(periphery)*: proved with Mathlib, whose axioms are reported.
- **How far seen.** *Theorem*: proved in Lean as stated. *Witness*: one constructed model, showing that something can happen, not that it does. *Reading* [AI reading, Dn]: an interpretation registered with its argument in the decision register; it changes no theorem. *Assessment*: a scoped evaluation with evidence and no Lean. *Thesis Tn*: an empirical claim, with how far it has been seen, what it forbids and the evidence so far.
- **Ranking results.** Where a result ranks courses, its theorem states the hypotheses it takes: the continuation condition (whose losses count), self-concern (a life's order over courses ranking above what it widens every widening that adds to its own continuation, read as what lies within its reach), `Comparison` (C, the requirement that an order rank each course above every course it widens, every claim counted as living) or the priority of basic conditions (J1/J2). Widening, one course keeping all another keeps and letting more form, is on a finite listing what every additive weighting that counts each listed item above zero ranks higher. The ethics, which emerges from continuation, is offered for life as a way of seeing, not a verdict; what of it is shown, what offered and what open, the philosophy dossier sets out (P-04). No item here uses these hypotheses.

Part 1. What the theory claims of the sciences

S-01 The verdict: restated or near-definitional

Meaning. An AI review placed each central result against its closest prior result. The rows bearing on the sciences, with their verdicts:

Result	Closest prior result	Verdict	Item
Coming and ceasing keep pace	"demographic balancing equation; trace theory (Mazurkiewicz 1977)" (ch. 8)	"near-definitional" (ch. 8)	S-07
Recurrence and attraction	"finite Poincaré recurrence; Landauer (1961); Bennett (1973)" (ch. 8)	"restated, without decidable equality" (ch. 8)	S-05
Rest needs a least loss	"ranking functions (Floyd 1967); Lyapunov; LaSalle; chip-firing" (ch. 8)	"restated" (ch. 8)	S-06
Fed from beyond, entropy, nets	"Schrödinger (1944); Prigogine (1977); ranking functions; P-invariants of Petri nets (Lautenbach 1975)" (ch. 8)	"restated; the nets agree with net theory and are not independent evidence" (ch. 8)	S-04, S-06
Loops	"nilpotent adjacency; Perron–Frobenius; Maruyama (1963)" (ch. 8)	"restated over whole amounts" (ch. 8)	S-10
Joint excess	"tragedy of the commons; positional goods (Hirsch 1976)" (ch. 8)	"near-definitional" (ch. 8)	S-11
Closures and selection	"Blackwell (1953); Pearl (2000); the proxy problem in its blindness to what a proxy does not register (Zhuang and Hadfield-Menell 2020)" (ch. 8)	"restated: their Theorem 1 on natural-number amounts, at optima rather than limits, with shared units for room; a discrete analogue at optima of their Theorem 2's first direction, with the responsive closure, pools and a re-reading's deadline" (ch. 8)	S-20
Evolution on descent, what bounds offspring numbers, Hamilton's rule and shaping among groups, groups on coupled conditions	"Price (1970, 1972); Fisher (1930); Hamilton (1964, 1975); multilevel selection; Malthus (1798)" (ch. 8)	"restated in counts; joined to the flow, a third part known as the Malthusian bound" (ch. 8)	S-07, S-08, A-10
Drift, fixation, with selection, mutation, the critical mass drawn	"Wright–Fisher; Moran (1958); Kimura; mutation–selection balance (Haldane); the error threshold (Eigen 1971); the Moran process of a game (Nowak and others 2004); stochastic stability (Kandori, Mailath and Rob 1993; Young 1993)" (ch. 8)	"restated as finite weighted bounds; risk dominance in counts, and a witness where the Moran criterion parts from it" (ch. 8)	S-07, A-10b

Since 3 November 2026, after edition 1, the chapter's table has a last column: "The last column lists, for each row, what the closest prior frameworks state that the account does not yet, as of 3 November 2026: stronger theorems, quantities, continuous and stochastic forms, wider scope." (ch. 8) "Four AI readers drafted it from the literature they knew, unless a linked record names the gap, and four others searched the repository to refute each item" (ch. 8), and "three further readers checked the column as written" (ch. 8); each entry "is a limit that holds so far and lapses when the account states the result" (ch. 8) (08 §8).

Beyond the table the review places regulation, the margin of a regime, the least loss from physics, the counted lock, conditioning, the exit from a margin, eventual widening, orders with a set point, the pace of selection and what keeping widening needs; each item below quotes its verdict. Of what the review found new, 08 §8 lists three things; one is "one vocabulary for conservation, selection limited by observation, and comparison of opportunity sets, which it calls a framing." (ch. 8) The other two, the premise ledger and the clarified ethics, are not this dossier's. The table's last two rows are not the review's: they "carry the verdicts of the audit of the third parts" (ch. 8), save what AJ8 added on 3 November 2026: the evolution row's shaping among groups, and in the drift row the critical mass drawn, its credits and its verdict's clause on risk dominance and the Moran criterion.

How far seen, and use. Assessment. "The verdicts come from an AI review that read each Lean statement against the literature it knew" (ch. 8). Every item below starts from its row.

Known weaknesses.

- "the search was not exhaustive" (ch. 8).
- The audit found: "The evolution line (Price, Fisher, Hamilton, drift, fixation, mutation) has no row in 08 §8's table, though 04 §7 points there for its verdicts." (a study record) The table has had the two rows since 27 October 2026; the line was never ranked by the review the other rows come from.

Questions.

1. For the rows of your field, is the verdict right, and does the chapter's last column name what your field states that the account does not yet? Is anything called restated in fact different from the known result, or anything called near-definitional in fact substantive?
2. Is any known result restated without credit?
3. Is one vocabulary for conservation, selection limited by observation and comparison of opportunity sets of use in your field, or a translation only?

S-02 Third parts: farther than a few frameworks, not than the sciences together

Meaning. "A containment of a framework has a third part where the theory forbids something the framework leaves open. Of the 21 rows the audit read, four have one, three of them distinct:" (ch. 8) how a part changes what can form, weakly (07; not treated here); joint excess under the gate (S-11); and the optimum laws (S-13). "Price's equation and Fisher's theorem joined to the flow have one as well" (ch. 8) (S-08), and the pace of selection a weak one (S-09); "drift, fixation and mutation have none." (ch. 8) The free-energy principle's containment "has five, two of them weak" (ch. 8) (S-17), joining claims of its literature, read as guarantees or unqualified, to other laws, and "Each is known (Curie 1894; the commons, with pivotality; identification, Rothenberg 1971; conventions symmetry does not select, Lewis 1969)." (ch. 8) Of the rest, the shape: "Each joins a framework that lacks conservation, renewal over time or a threshold to a premise or law of the theory, and forbids what the conjunction forbids. ... Each

forbidden statement of the other rows is known in a neighbouring field (critical depensation, the Malthusian bound, bioeconomics, scheduling), and no forbidden statement, the principle's included, is one a specialist in the field where it is known would not foresee." (ch. 8) The verdict: "So the theory reaches farther than the few contained frameworks that lack conservation, renewal over time or a threshold, by joining them to its own premises and laws." (ch. 8) Of the principle, "the theory states the central quantities on its own models; it reaches farther than the principle alone in what it forbids, the five statements above, and states less than it where it states nothing yet, in densities, continuous time and deep models among others" (ch. 8) (S-17). "With the rest (thermodynamics, recurrence, control, constructor theory, causal sets among them) it is level or states less, and it does not yet reach farther than the sciences taken together" (ch. 8)

How far seen, and use. Assessment: an AI audit (UE0) and an AI search in three rounds (UE6). Of the search: "found no result that no neighbouring field states" (ch. 8). What the audit counts as the theory's own: "Within the third parts, the generality is the theory's: every order rising with what couples in, every relation, every renewal law bounded by `r`. So are the exactness (necessary and sufficient, with witnesses on both sides) and room derived from one step of exchange. These make the joins stringent in the plan's sense of patterns that constrain what may be filled in. They do not make the forbidden statements new." (a study record)

Known weaknesses.

- "The literature verdicts come from the AI's prior knowledge." (a study record) They also draw on the repository's own prior-art checks of 20–23 October, "which checked several sources on the web; this audit made no new web search" (a study record).
- "A bounded search, not established novelty and not established absence: three rounds, each by a different reader, each with a literature search" (a study record).
- [this dossier, AI] Price's equation and Fisher's theorem in its exact case are identities, which forbid nothing. A third part found by joining them to conservation is the conservation premise's, so the bar it clears is low.

Questions.

1. Is each forbidden statement known in your field, as the audit says, and where is it stated?
2. Are the joins to conservation, renewal or a threshold already made in your field's own models, so that the theory's form adds nothing?
3. Is a third part, so graded, the right test of whether a containment reaches farther?

S-03 Strength as content: the theses and the unifications

Meaning. "Strength is the observations a claim excludes; how far a claim has been seen is whether it is proved under named premises, supported by frozen tests, conjectured or argued; the two are kept apart." (ch. 8) Each thesis states what it forbids and its evidence. Where each is treated:

- T1 (persistence certifies the joint draw): S-14. T2 (the state of a renewable condition): S-15. T6 (a break is seen coming only through what couples into what breaks): S-16.
- T3 (optimization drains what lies beyond its couplings, where that gives room): its law for optima, S-13 and S-20; its clause on trained systems, the alignment dossier (A-18).

- T7 (a measure read from what it selects outlives its refutation): the alignment dossier (A-08, A-19). T4, T5 and T8, the ethics' theses: the philosophy dossier (P-26); T4's test IN1 is in A-18.

Four unifications: "Four results known in substance are each claimed as one law across fields that state them apart." (ch. 8) Supervisory control as one selection (S-20); Landauer's and Bennett's irreversibility as one merge (S-05); Zhuang and Hadfield-Menell's Theorem 1 as one optimum law (S-13); and the monotonicity of opportunity sets as widening, what every additive positive weighting shares on a finite listing, the philosophy dossier's (P-07). How far they have been seen: "The literature here is from the AI's knowledge unless a linked check read it." (ch. 8) "Read as a correspondence, a unification adds no theorem and changes no verdict of" (ch. 8) §8's table. "Read as a thesis, it forbids what the source fields leave open." (ch. 8)

How far seen, and use. A method. "A thesis adds no theorem, a refutation removes none, and a failure stays recorded beside the thesis." (ch. 8) And of the frozen tests: "A frozen rule fixes how a verdict is scored before the data, not that the rule reads the construct the verdict names" (ch. 8). "So a verdict is read against every measure of its construct the run produced, and where they disagree with the rule's print, the disagreement is reported and the verdict not claimed." (ch. 8) (The rule's print is the verdict the frozen rule outputs.)

Known weaknesses.

- [this dossier, AI] T1, T2 and T6 are stated for every renewable or drawn condition, and their frozen tests cover one to three systems each. How far each has been seen says so, but the theses are much wider than their evidence.

Questions.

1. Are the forbidden lists of T1, T2 and T6 specific enough for data in your field to refute them?
2. Is keeping how far a claim has been seen apart from its strength useful to you as a reader of the theses?

Part 2. Physics

S-04 Balance and the least loss, read from thermodynamics

Meaning. Periphery theorems (in `physics/EntropyProduction.lean`) read World premises from thermodynamics. A level is read as whole units of available work (exergy) against one reference environment.

- *Balance from thermodynamics' first law.* A quantity carried only by the exchanges taking part balances the levels exactly where no condition produces it (`balances_iff_unproduced`), so a locally conserved quantity such as energy balances (`balances_of_conserved`); "conservation of the total alone does not, as a unit moving between two conditions with no exchange shows" (ch. 1). With thermodynamics' first law and the entropy balance at each store and exchange, and levels read as available work, one reading gives balance exactly where no store generates entropy, no gain where no exchange generates negative

entropy, and dissipation where, besides, every exchange taking from a closed family generates some (`balance_and_least_loss`).

- *Least loss from entropy production.* "Under the exergy balance and the second law nothing comes from nothing at the unit: the second law in available work gives the spine's first law, and destroyed exergy is its record." (ch. 2) (`EntropyProduction.gainless_of_secondLaw`.) "The second law alone gives no dissipation" (ch. 2) (`EntropyProduction.secondLaw_not_dissipative`). Irreversibility of every taking with whole units gives it at that unit, though not at every finer one (Known weaknesses), and so does "a floor of destroyed exergy at least the unit, at every finer unit" (ch. 2) (`EntropyProduction.dissipative_of_floor`). "Landauer's bound is no such floor by itself: it bounds the heat an erasure exports, heat exported is not exergy destroyed, and a logically irreversible step can be carried out thermodynamically reversibly (Bennett 2003; Sagawa 2014)." (ch. 2) For one family, balanced on each run, with a finite measure and the family's stock, uptake, replenishment and takers integrable at each step: if from some step on the family's expected replenishment and expected takers together are at most its expected uptake (dissipation in expectation), then almost surely nothing takes from the family from some step on (`EntropyProduction.taking_not_recurring_ae`). The integral fluctuation theorem, taken as a hypothesis, bounds the probability of a gain (`EntropyProduction.gain_probability_le`).

How far seen, and use. Theorems (periphery, with the axioms of Mathlib's reals). "Energies and entropies are reals, levels whole units of available work, and the step from physics is a stated correspondence" (ch. 1). "Every least loss that physics supplies is a theorem of stochastic thermodynamics, with premises of its own." (ch. 8) The audit's verdict: "a premise located, not a result" (a study record). The physics the record reads: "Gouy–Stodola: exergy destroyed is T0 times the entropy generated, never negative" (a result record); "Kelvin–Planck says no exchange draws that record back." (a result record) "Landauer (confirmed by Bérut et al. 2012 and Jun, Gavrilov and Bechhoefer 2014; its general derivation disputed by Norton)" (a result record). "Fluctuation theorems (Jarzynski, Crooks, Seifert; observed by Wang et al. 2002 and Collin et al. 2005): single small runs gain, so the premise there is dissipation in expectation." (a result record) Two of the theorems are known results: "The probability bound on a gain is the integral fluctuation theorem's Markov bound." (a study record) The almost-sure law is a Borel–Cantelli argument, and its proof uses Mathlib's Borel–Cantelli lemma.

Known weaknesses.

- "That strict irreversibility gives a least loss at a refining unit: a strict loss without a floor can fall below every unit; only a floor in energy or a proportion survives refinement." (a result record)
- "The integral fluctuation theorem itself, or local detailed balance, from which stochastic thermodynamics derives it; both are hypotheses here." (a result record)
- "That balance is a theorem of the dynamics: it is relocated to thermodynamics' first law with locality, which stays a premise about what is represented." (a result record) And "the pressure and chemical terms of available work are not represented." (a result record)
- Of the chapter's example, a later AI search found "Landauer's bound alone does not supply the World's least loss" (a study record) because "heat exported is not exergy destroyed (Sagawa 2014)" (a study record); the chapter now says so (Meaning), and a floor at every finer unit stays a premise physics does not supply by that bound.

- [this dossier, AI] The theory's *first law* (`Flow.Gainless`: across every step, no formation gives a closed family without repetition more than it takes from it) is given by the second law read in available work, while 01's balance theorem reads thermodynamics' first law. The account names the two apart, the spine's first law and thermodynamics' first law in full, but a physicist may still find the name misleading.

Questions.

1. Is reading a level as whole units of exergy against one reference environment a fair correspondence, and what does it lose?
2. Is it right that every physical floor on dissipation (a logical merge, finite time, a finite bath) is a theorem of stochastic thermodynamics with premises of its own, and could the World's least loss be derived from a weaker one?

S-05 Recurrence, merges and Landauer's count

```
theorem Lossless.reset_unlisted {next : S → S} (lossless : Lossless next) {part : S → P} {p : P}
    (reset : ∀ x, part (next x) = p) {y : S} (other : part y ≠ p) :
    ¬ ∃ all : List S, ∀ s, s ∈ all
```

File: `Becoming/LeastLoss.lean`.

Meaning. (any type)

- *Recurrence.* "Within finitely many configurations, a list holding every one, a determined law merges nothing exactly where every configuration returns, exactly where it keeps every possibility" (ch. 1) (`Law.lossless_iff_returns`, `Law.returns_iff_keeps`). Within them, a quantity that never falls along an orbit and rises at some step of it rules out a law that merges nothing (`Law.rise_not_lossless`).
- *A part kept by the rest* (the block). `Lossless next` says the law `next` takes no two configurations to one. If such a law sets a part to one value `p` at every step, and some configuration's part is not `p`, then no list holds every configuration: resetting a part without merging needs unbounded room in the rest. In the witness `Reset`, a one-bit part is set to false at every step and the rest keeps the history of what it held (`Reset.kept_by_rest`). Narrowing the part spreads the rest (`Law.Lossless.narrowing_spreads`), and a kept history never returns (`Law.withHistory_never_returns`).

How far seen, and use. Theorems. "This is Landauer's (1961) count with Bennett's (1973) history; no energy, temperature or probability is read, so kT ln 2 is not derived" (ch. 2). The third unification reads one merge across computation, heat, extinction and forgetting, with how far it has been seen, in 08's words: "the law proved for determined maps on finitely many listed configurations; its reading across computation, heat, extinction and forgetting a correspondence; kept as a law, since as a thesis it forbids little" (ch. 8).

Known weaknesses.

- Its record's correspondence: "The law is that phase-space count, with Bennett's history (1973) as its witness." (a result record) And "the step from a count to heat (S = k ln W, heat at least T times the entropy a bath at temperature T receives) stays a physical premise." (a result record)

- The record credits later work as well, from Bennett's notes of 2003 (SHPMP 34), of which it quotes "in a sense ... a restatement of the Second Law" (a result record), to "Ladyman, Presnell, Short and Groisman (SHPMP 38, 2007; disputed by Norton). The finite-rest floor is the counting shadow of Reeb and Wolf's finite-bath correction (NJP 16, 2014) and Masanes and Oppenheim's unattainability (Nat. Commun. 8, 2017)." (a result record) UE6's first round adds "Reeb and Wolf 2014 with Sagawa 2014 (Landauer's bound, and why logical irreversibility is not exergy destruction)" (a study record).
- "The list is the hypothesis each law writes in; no unfolding is assumed to have one, and no decidability is used" (ch. 1).
- [this dossier, AI] Landauer's argument concerns phase-space volume under Hamiltonian dynamics; the law counts configurations of a finite deterministic map, so the correspondence carries the physics only through the count-to-heat premise.

Questions.

1. Is a deterministic map on finitely many listed configurations an adequate model of a logically irreversible step?
2. Does reading extinction and forgetting as merges add anything beyond analogy?
3. Beyond Landauer (1961), Bennett (1973) and the later work the record and the search name (Known weaknesses), is anything in the thermodynamics of computation missing from the credit?

S-06 Rest needs a least loss; continuing change is fed from beyond

```
theorem fed_from_beyond [DecidableEq C] (balances : L.Balances W) (dissipative : L.Dissipative)
    {ds : List C} (distinct : Distinct ds) {T₀ : Nat} (around : ∀ t, T₀ ≤ t → L.Around ds t)
    (recurs : ∀ N, ∃ t, N ≤ t ∧ 0 < L.takers ds t) (N : Nat) :
    ¬ ∀ t, N ≤ t → ¬ L.FedFromBeyond ds t
```

File: `Becoming/Continuation.lean`.

Meaning.

- *Rest* (any type; World for the instance). "A closed amount that keeps losing need not stop: a stock halving at every step never rests. Rest needs a least loss at each change, or the amount read off finitely many configurations." (ch. 2) (`Amounts.rests`; `Amounts.rests_of_finite_reading`, with no least loss; `Halving.stock_unlisted`.)
- *The rest theorem* (World). With censuses of what is active and of what stands, arising embodied, balance, dissipation, and every step from some step on enclosed around what stands, rest from some step on cannot be refuted (`World.comes_to_rest_unrefuted`). Read the other way, with a census of what is active, every relatum embodied, balance and dissipation, arising without end leaves no step from which every step is enclosed (`World.arising_opens`). Each census and full embodiment are needed: "with only arising embodied, a world can be enclosed at every step and arise without end" (ch. 2) (`World.rest_needs_standing`: its first world has a census of what is active and none of what stands, its second every relatum embodied and no census of what is active; for the rest theorem's census of what is active, `World.rest_needs_census`). Openness does not give arising without end (`World.holding_world`).

- *Fed from beyond* (World; the block). Under balance and dissipation, with equality of conditions decided, take a finite family without repetition, enclosed around from step `T₀` on and taken from at arbitrarily late steps. It cannot be that from some step `N` on no formation takes from outside the family and gives to a member.

How far seen, and use. Theorems. Closest prior results for the last: "Schrödinger (1944); Prigogine (1977); ranking functions; P-invariants of Petri nets (Lautenbach 1975)" (ch. 8). "The theorem says rest cannot be refuted, not when it comes" (ch. 2). Of feeding: "no step of feeding is named" (ch. 2).

Known weaknesses.

- "It needs balance at a fixed unit of natural-number amounts, each taking step losing at least one whole unit of it" (ch. 2). "at a unit that splits at arbitrarily late steps the conclusion can fail" (ch. 2). Since 3 November 2026, after edition 1, where edition 1 quoted that with real amounts the law fails: "its descent is the whole-number case of a law over any Archimedean amounts with a least loss above nothing" (ch. 2); "No theorem states the law over real amounts" (ch. 2).
- "The exclusion, that an enclosed family taken from without end cannot go unfed from beyond, is the sources' own. The theory forbids no more and needs more to forbid it." (a study record)
- [this dossier, AI] The law gives no amount, rate or time of feeding, which an entropy-production balance would give.

Questions.

1. Is a least loss of one whole unit per taking step, at a fixed unit, a fair idealization of dissipation, given that the conclusion can fail at a unit that splits without end?
2. Do the closest prior results quoted above already forbid what this law forbids?
3. Is a law that says only that rest cannot be refuted, not when it comes, of use in your field?

Part 3. Evolution

S-07 The census and the classical laws in counts

Meaning.

- *Coming and ceasing keep pace* (World). "Over any stretch an amount at the end with all it fell is the amount at the start with all it rose" (ch. 1). For a decided determination over a finite family, coming to be recurs exactly when ceasing does (`coming_recurs_iff`).
- *The classical laws in counts* (any type). Price's equation in counting form, with no premise (`Becoming.price`). Fisher's theorem in its exact case: with exact transmission the mean offspring number rises exactly where offspring numbers vary (`Becoming.Reproduction.Generations.ByNumber.rises_exact_iff`); with breeding values that meet the normal equations, transmitted exactly, the mean breeding value rises by the additive variance over the mean offspring number (`Becoming.Reproduction.Breeding.response`). Drift has no direction, and the histories fixing a kind are bounded on both sides, in finite weights, by its starting share (`Becoming.Drift.fixation`); with selection in the draw, by the ratio that is the Moran probability in

counts, in the same finite form (`Becoming.Drift.Replacement.fixation`), and, with weights read from the count and a weight of staying that makes every step weigh T, wherever each up is at least a positive c (`Becoming.Drift.Replacement.fixationBy`, in 04's text since 3 November 2026, after edition 1); mutation–selection balance and Eigen's (1971) error threshold (`Becoming.Reproduction.Mutation.kept_share`, `Mutation.falls_of_outrun`). Hamilton's rule is stated in its general additive form on any list, which needs no groups: where each member's weight with k per unit of its own amount is a base with b per unit of a reading, the whole selects for the amount exactly where b times the reading's covariation with it exceeds k times its spread (`Becoming.SelectionLevels.rises_iff_reading`), so, with a member's group total as the reading and k = b + c, "the whole's verdict through that covariation, though not through the spread among groups, holds for groups of any sizes" (ch. 4) (in 04's text since 3 November 2026, after edition 1). It is stated in groups of equal size with additive returns (04), and since 4 November 2026 at any sizes, where each unit a member keeps costs it K, brings s back to it alone and gives q to every member of its group: each member gaining for every unit its group holds, the verdict goes through the covariance, member by member, of a member's group total with its group's mean (`Becoming.SelectionLevels.rises_iff_kept_sized`), at unequal sizes not a function of the whole's spread and the size-weighted spread of the means, two listings with the same spreads, one step with the groups listed as given and their sizes fixed over it, selecting for keeping and against it (`Becoming.SelectionLevels.Sized.either_verdict`, 04); what each unit kept gives its group divided among the members instead, it goes through the size-weighted spread of the means: over the rationals, where the amount varies, $K < s + Q\rho$, with Q what a unit gives its group and ρ the share of the variance, weighted by size, lying among the means (`Becoming.SelectionLevels.rises_iff_divided_sized`); "which sharing actual groups have, which is a premise" (ch. 4); the multilevel split weighted by size is the general law (`Becoming.SelectionLevels.rises_iff_sized`, 04); and, on the World with balance on the family and the coupled gate, groups on coupled conditions (04).

- *The selection term and a lineage's fate* (any type; World for lineages). "The selection term proper reads how offspring numbers differ, not how many there are" (ch. 4). So within any horizon a course nonempty at it has its cohorts and terms through it shared by a course nonempty at every cohort and by one empty from two steps after it on (`Horizon.two_fates_of_nonempty`), and from a nonempty first cohort, terms zero from some step on are shared by a course that goes on and one that ends (`Horizon.eventually_zero_two_fates`). Terms positive at unboundedly many steps keep every cohort nonempty (`Generations.never_empties_of_unbounded`). "Which side a course is on is not decided." (ch. 4) The witness of evolutionary suicide's pattern, reproduction simply stopped and the mark a label rather than a draw, has a term positive at every nonempty cohort but the last, where it is zero, while every member degenerates, everything continuing it ending (`Courses.positive_but_last`).

How far seen, and use. Theorems. "The laws of §3–§6 restate known results" (ch. 4) and "It adds no new population-genetic prediction." (ch. 4) The audit: "counting forms of the classical results" (a study record).

Known weaknesses.

- Since 3 November 2026, after edition 1, which listed it as not stated: "Fisher's identity itself, through one carried amount or several, is stated with ... cross-multiplied" (ch. 4).

- Of breeding values: "That the normal equations have such a solution is given, not shown" (ch. 4), and "inexact transmission of the breeding value is not stated" (ch. 4).
- The method chapter lists among prose that once said more than the theorems "that the census is a law of how anything comes to be in every universe" (ch. 8).
- The limit is read from finite bounds, not proved: "finite bounds, whose reading is that the fixing share tends to h/N, the neutral fixation probability." (a result record) The chapter likewise: "finite weighted bounds, whose reading is that the fixing share tends to" (ch. 4).

Questions.

1. What, if anything, do these laws add to the classical results they restate in counts?
2. Does stating these laws in whole numbers, with offspring lists supplied and fitness not defined, lose anything that matters?
3. The table gives the evolution line the audit's closest prior results and verdict, which the review behind its other rows did not rank (S-01). Would you give the same?

S-08 What bounds offspring numbers: Price and Fisher joined to the flow

```
theorem closed_lineage_lowered {ds : List X} (descended : SelectionTerm.Flow.Descended L W G)
    (within : ∀ t p k, p ∈ G.family t → k ∈ G.kids t p → k ∈ ds)
    (balances : ∀ t, t < W.stock ds 0 + 1 → L.BalancesOn W ds t)
    (dissipates : ∀ t, t < W.stock ds 0 + 1 → L.DissipatesOn ds t)
    (number : Generations.ByNumber G) (reproducing : 0 < G.size 1) :
    ∃ t, t < W.stock ds 0 ∧ G.concordAt t + G.size t * G.raisedAt t < G.size t * G.loweredAt t
```

File: `Becoming/Reproduction.lean`.

Meaning. (any type; World for the bounds) Each offspring is given at least one unit. So a parent's offspring number is at most what it gives them (`Becoming.Reproduction.Flow.born_le_given`), with its distinct offspring standing within a family dissipated on, and one more where there is any, at most what it takes from the family (`Flow.number_within_take`), and across a step at which a family is balanced and dissipated on, distinct offspring standing within it number at most its stock (`Flow.born_within_stock`). Over cohorts of descent within a family balanced and dissipated on at each step, a lineage reproduces at most the starting stock many steps (`Flow.reproducing_within`) and then ends (`Becoming.Reproduction.Flow.lineage_ends`). The block reads this by offspring number. A lineage that reproduced at its first step within such a family has a step, before the stock is spent, at which transmission lowered offspring number by more than selection and what transmission raised: in the block, `G.concordAt t` is selection on offspring number (zero exactly where offspring numbers are uniform), and `G.raisedAt t` and `G.loweredAt t` are what transmission raised and lowered. Witness: without dissipation a lossless relay keeps a lineage going within a family of stock one (`Becoming.Reproduction.Relay.relay`).

How far seen, and use. Theorems. "offspring numbers are bounded by what the flow gives and takes, under the stated descent and resource hypotheses, though the offspring lists themselves stay supplied" (ch. 4). A third part, in the audit's words: "in a family closed to inflow and dissipated on, a lineage reproduces for at most as many steps as the family's stock holds units, and Price's transmission term must at some step lower offspring number by more than selection raises it" (a study record).

Known weaknesses.

- "Whether actual lineages lie within such families is an application question." (a result record)
- "That a finite closed resource ends reproduction is the Malthusian bound." (a result record)
- [this dossier, AI] The bound rests on each offspring being given at least one whole unit. With real amounts and no floor on what an offspring is given, offspring number is not bounded by the stock.

Questions.

1. Do resource-explicit models of reproduction, such as consumer–resource or chemostat models, already state this law?
2. Is an offspring given at least one unit of a conserved amount a fair model of the cost of reproduction?
3. Does joining Price's equation to a conserved flow give population genetics anything it lacks?

S-09 The pace of selection within what flows in

```
theorem whole_within_power {G : Generations X} {m k T : Nat} (positive : 0 < m)
    (balanced : ∀ t, t < T → L.BalancesOn W (lineage (G.family t) (G.kids t)) t)
    (arise : ∀ t p c, t < T → p ∈ G.family t → c ∈ G.kids t p → W.ComesToStand t c)
    (stands : ∀ t p c, t < T → p ∈ G.family t → c ∈ G.kids t p → m ≤ W.level (t + 1) c)
    (exact : ∀ t, t < T → Exact (G.family t) (G.kids t) (G.amount t) (G.amount (t + 1)))
    (capped : ∀ t p, t < T → p ∈ G.family t → W.level t p ≤ m)
    (fed : ∀ t p, t < T → p ∈ G.family t → L.inflow (lineage [p] (G.kids t)) t + m ≤ k * m)
    (sustained : G.size 0 ≤ G.size T) (whole : G.size T ≤ G.sum T) :
    G.size 0 ≤ k ^ T * G.sum 0
```

File: `Becoming/SelectionPace.lean`.

Meaning. (World) Across a step at which the flow balances on the family a lineage occupies, each offspring coming to stand at `m` or more, `m` times a parent's offspring number is at most what the parent stood at and what flows into its lineage (`SelectionPace.parent_within_inflow`). With exact transmission of the amount carried (each offspring carrying its parent's amount) and parents standing at no more than `m`, in a population not shrinking `m` times the selection term is at most the size times the inflow into the carrying lineages (`SelectionPace.selection_within_carried`); with nothing flowing into them, a rise of the mean shrinks the population (`SelectionPace.unfed_rise_shrinks`). The block is the per-parent form. Where what flows into each parent's lineage, with `m`, is at most `k · m`, then in a population not shrinking (`sustained`), reaching the whole (`whole`, the final total at least the final size: for a 0/1 amount, every member carrying it at `T`) from `c` carriers among `N` needs `N ≤ k ^ T · c`, so one substitution takes at least `log_k (N / c)` steps. With offspring number itself inherited, the squared response of any amount is bounded by the next step's inflow and the amount's spread (`SelectionPace.response_sq_within_inflow`), and not with the same step's inflow (`SelectionPace.Inherited.same_step_fails`).

How far seen, and use. Theorems. "so the third part against Price and Fisher is yes and weak, carried by its premises." (ch. 8) Its neighbours: "the form behind the reproductive-excess limit of Nei (1971) and Felsenstein (1971), and not the spacing of successive substitutions that limit states. It is not Haldane's cost (1957), which counts selective deaths the model does not have." (ch. 8) The squared response is "the bound of the selection differential by the opportunity for selection (Crow 1958; Arnold and Wade 1984), with the opportunity bounded by the inflow" (ch. 8).

Known weaknesses.

- "The natural statement, a pace bounded by what flows into the whole, is false: a part of a closed whole rises by drawing on the rest." (ch. 8) (`SelectionPace.Displacement.natural_fails`.)
- "That actual populations meet exact transmission of offspring number or the per-member bounds; soft selection is a reading of Displacement, not a theorem." (a result record) And "A least standing from the dynamics: m is a supplied level, tied to the web's need only through stands_of_need." (a result record)
- "The third-part verdict, yes and weak, stands. Malthus is the main neighbour" (a study record).

Questions.

1. Does the per-parent bound capture the reproductive-excess limit, and is it told apart from Haldane's cost correctly?
2. Is the squared-response law a correct reading of the opportunity for selection?
3. Do any natural populations meet exact transmission of the amount carried and parents standing at no more than `m`, or, for the squared-response law, exact transmission of offspring number, even approximately?

S-10 Loops and regulation

Meaning. (World) A loop that forms at a step passes through a coupling acting at that step (`Closing.forms_by_closing`); before a loop forms, the amounts need not show whether one will: in a witness, a loop forming at a step and one forming later carry the same amounts up to the first (`Closing.Onset.unforeseen`). "With whole-number factors a loop closes at once, keeps, or exceeds every bound, never closing gradually" (ch. 2) (`LoopReturn.nothing_between`). Under the standard bounds the difference carried to a step is at most the starting gap times `(1 + M)` to the power of the number of earlier steps at which anything couples, `M` bounding the strengths into each condition, and a self-loop attains it (`carried_le_count`). A law reading a stock only below a threshold couples it with strength zero above it, so the distance to the break is carried nowhere (`break_carried_nowhere`).

Regulation (any type; World). "A rule reading `ℓ` steps late keeps no band `[a, b]` with `0 < a` narrower than `2(ℓ + 1)d` against rests moving the level by at most `d` a step, attained by the late return (`Regulation.no_narrower_band`)." (ch. 2) With balance, where the joint exchange gives no more than it takes below `m`, a level below `m` never returns to it, gated renewal being the case (`Flow.never_returns_below`); where at each condition of a family the joint exchange stays within the gap to that condition's own level and opposes departure from it from some step on, the family's stock settles at the total of those levels, nothing reading the total (`Flow.stock_settles`).

How far seen, and use. Theorems. Closest prior results for the loops: "nilpotent adjacency; Perron–Frobenius; Maruyama (1963)" (ch. 8); verdict, "restated over whole amounts" (ch. 8). A calibration: "That every schedule settles, each coupling recurring or stopping, is exactly the dichotomy `RecursOrStops`, which gives LPO (`settled_iff_recursOrStops`): settling is deciding whether a coupling acts again." (ch. 2) For regulation: "Regulation restates deadbeat control and requisite variety (Dahleh and Pearson 1987; Ashby 1956)." (ch. 8) The late reading's band was tested with learners in the gated commons (CG1): its predictions E1 and E2 are of the identity class, and E3, a learned adversary breaking a band one narrower

against the late return and a learned overseer, passed in 80 of 80 seed-cells: "The late reading's bound is realised by learners." (the experiment's README)

Known weaknesses.

- "excludes gradual closing only because factors are whole numbers. Real factors close gradually; this is an artefact of the commitment, not an exclusion of a real course." (a study record)
- The method chapter lists among prose that said more than the theorems "that nothing a part senses shows how near a break is. The theorem assumes the stock is not sensed, and the early-warning literature (Scheffer and others 2009) shows the general claim false" (ch. 8); `break_carried_nowhere` invites the same reading (S-16).
- Of regulation, what the record does not show: "That a given system regulates a level, which level, how late it reads, or the size of its step." (a result record) Nor "One policy keeping a band from every start in it under late reading; several regulated levels lumped into one; regulation of several conditions at once." (a result record)
- Of CG1's E rows: "E is the law's own one-condition setting, not the commons' dynamics." (the experiment's README)

Questions.

1. What, if anything, do these laws add to the combinatorics of walks on a weighted graph over whole amounts?
2. Is the law that a stock read only below a threshold carries the distance to its break nowhere known for piecewise-smooth or hybrid systems?
3. For control theory, does the late reading's band add anything to deadbeat control and requisite variety?

Part 4. Ecology and renewable resources

S-11 Joint excess under a gated renewal

```
theorem joint_excess_ends (balances : L.Balances W) (acting : L.Acting W)
    (dependent : W.Dependent) {c : C} {m : Nat}
    (gated : ∀ t f, f ∈ L.present t → 0 < L.give t f c →
      W.requires t f c ∧ m ≤ W.need t f c)
    {T n : Nat} (excess : ∀ k, k < n → L.given (T + k) c < L.taken (T + k) c)
    (start : W.level T c < n + m) :
    ∀ k g, W.requires (T + n + k) g c → m ≤ W.need (T + n + k) g c →
      ¬ W.active (T + n + k) g
```

File: `Becoming/Continuation.lean`.

Meaning. (World: balance, only active giving, activity needing support; the gate local) Formations can each take within what a step gives while together they take more (`Pair.own_not_joint`). Each step of joint excess lowers a balanced level by at least one. The block: where every formation giving to condition `c` depends on it at a need of at least `m` (`gated`), `n` consecutive steps of joint excess from a level below `n + m` leave inactive, at every later step, every formation that then requires `c` at a need of at least `m`. "In a

minimal witness two takers end themselves and their renewer, one taker does not." (ch. 5) *The rest outruns the renewal*: where the others alone, listed without repetition, take more than a step gives at each of `n` steps from below `n + m`, every formation depending at that need ends whatever one member takes (`Flow.outrun_ends`), and, at a need of one, every score reading the condition only from the rest's end on scores the member's acts alike (`Flow.outrun_reading_alike`): "no weight on such a reading selects the member's restraint" (ch. 5).

How far seen, and use. Theorems; a third part (S-02). "It restates the tragedy of the commons and overshoot (Lloyd 1833; Hardin 1968; Meadows and others 1972) in near-definitional arithmetic, not who ought to take less" (ch. 5). Tested with learners in a gated commons (CG1) and with the rest's draw varied (CG2), whose learner results the alignment dossier gives (A-18); learners that generalize across states, run on CG2's conditions, "kept the stock in three of the six locked settings (`k3s1_lam32`, `k3s8`, `k3s1_share`)" (ch. 7) (experiment). Of the identity predictions, "The identity class gives no empirical support to the theory beyond the correctness of its identification." (the experiment's README) Its scope prediction passed, the one the account calls informative: "Where two patches lie within the regrowth radius, joint excess on one patch read alone does not end its renewal (all 679 windows violated, 100 of 100 runs regrow after reaching 0). Read as one condition, the two patches meet the law." (the experiment's README) Of the outrun law with learners, "Count: of eleven conditions, nine passed and two failed; of five contrasts, three passed and two failed." (the experiment's README)

Known weaknesses.

- Corrected on 6 November 2026, after edition 3, which gave this limit with no step named: "The gated conclusions after the step the excess reaches are proved only for renewal depending on what it renews; a taker not so depending can continue" (ch. 5). "At that step the end holds for any renewal (`Flow.ends_at_race_end`), and a formation renewed from outside can be active again (`Restored.returns`)." (ch. 5) "Read on a lineage at the kind grain, where everything a line reaches at that step so depends, the line from before that step does not continue past it, so such a line drawn past renewal from outside, such as a sun's, from a stock below `n + m` degenerates with no gate" (ch. 5) (04 §4); "no such line crosses the step, which does not say that nothing of the kind stands after it." (ch. 5)
- "The forbidden statements are new to the row's sources and not to population ecology or threshold games" (a study record).
- "Stated for natural-number amounts; excess compares totals at each step and treats neither reserves nor delayed renewal." (a result record)
- "That depletion is the only equilibrium: where the rest outruns, restraint by all may also be one; what learners reach is a claim about learning, tested in CG2 and EM1, not a theorem." (a result record)
- [this dossier, AI] The gate puts the threshold into the premise, and, at a need above the least that can renew at all (the least of its kind that can breed under any conditions, apart from any level this stock's renewal needs), it is stronger than depensation as fisheries measure it. Under the gate nothing renews the condition below `m`: with nothing drawn a gated level below `m` stays where it is, and with a standing draw it falls as under critical depensation (growth below zero beneath a threshold). Depensation in Myers et al.'s sense is per-capita recruitment falling at low abundance while recruitment stays positive. So, at such a need, the gate is the limiting form of critical depensation in which renewal below the need is zero, and, on a stock's own renewal, how often depensation occurs bounds from above how often the gated conclusions after the step the excess reaches apply at such a

need. At a need of one, counted in the least that can renew, it bounds nothing: the gate then asks only that what gives depends on at least one unit of it, so that, with only active giving and activity needing support, a level of nothing in that count gets no renewal, which a stock renewed only by its own members meets with no depensation, and those conclusions say that such a stock, once drawn to nothing, stays there and what depends on it at a need of at least one ends. At the levels studied depensation has rarely been detected: significant depensation was found in 3 of 128 fish stocks (Myers et al. 1995), and a meta-analysis of 207 stocks of 104 species found strong compensation in many species, much weaker compensation than expected in others, and evidence of depensation in some, such as Atlantic cod and Alaskan walleye pollock (Keith and Hutchings 2012).

Questions.

1. In which systems is renewal gated at a need above the least that can renew at all, zero below it, and can `m` be measured in advance there?
2. Does the outrun law add anything to the theory of threshold public goods and dynamic commons games?
3. What are the clearest field cases of a renewal that depends on what it renews?

S-12 The margin of a regime

```
theorem ends_iff_below {g : Nat → Nat} {H y hi x : Nat} (pos : 0 < H + y) (short : Short g H y)
    (nothing : FromNothing g) (meets : Meets g H y hi) (keeps : Keeps g H hi) (high : x ≤ hi) :
    (∃ n, (Law.run (step g H) x).state n = 0) ↔ x < H + y
```

File: `Becoming/RegimeMargin.lean`.

Meaning. (the regime; any giving law, whole numbers) One condition under a standing draw `H` steps from `x` to `(x − H) + g(x − H)` (`RegimeMargin.step`). In the block, `Short g H y` says giving falls short of the draw below `y`; `FromNothing g` that nothing gives nothing; `Meets g H y hi` that giving meets the draw at every `z` from `y` with `z + H` at most a top `hi`; `Keeps g H hi` that the step keeps levels up to `hi` within it; and `0 < H + y` that the draw and `y` are not both zero. Then a level up to `hi` ends exactly where it starts below `H + y`: where that is at most `hi`, the point of no return is `H + y`, exact up to `hi` (`RegimeMargin.ends_iff_below`). Where giving stops below a need, the draw is positive and giving meets it at `y`, the point lies at least the draw above the need (`RegimeMargin.root_above_need`), and, `y` the least level meeting the draw, more than that exactly where giving at the need falls short of the draw (`RegimeMargin.beyond_draw_iff`). "Draws add, and their rises at the point are superadditive up to one" (ch. 2): with joint giving subadditive above the need and, at each amount left, no greater than under either draw, where the least levels meeting the two draws and the joint draw lie `a`, `b` and `c` above the need, `a + b ≤ c + 1`, so the joint draw's point, the draw above the least level meeting it, rises above the need by at least the sum of the two draws' rises above it less one (`RegimeMargin.rises_superadditive`).

Where a run turns (02, since 10 November 2026): in whole numbers, a run whose first step does not lower it and which later falls names two levels, the lower first, that the step carries the other way (`RunTurns.turns_names_levels`), and on the regime two amounts left, the less first, at which more left is given less (`RunTurns.regime_turn_overcompensates`). The chapter calls such a turn overcompensation; [this dossier, AI] the stronger of the two conclusions is the first, since more left given less also holds of a giving whose step still keeps the order. Where giving rises with what is left the run moves one way only. A

step keeping the order of the levels up to `hi` and keeping them within `hi` brings every run starting in the band to rest within `hi` steps (`BandRest.rests_on_band`); one reversing the order there and keeping the band, to rest or period two from step `hi + hi` (`RunTurns.reversing_period_two`), of which the record says "That the square of an order-reversing map keeps the order is a known fact" (a result record). A witness that a run on a band the step keeps need not end at rest or period two: under a draw of one, giving three from nothing and nothing from more, so outside the premise above that nothing gives from nothing, the run from one has period three and the step neither keeps nor reverses the order (`RunTurns.ThreeCycle.returns`, `RunTurns.ThreeCycle.neither`). None of this forces a turn, and the record credits the nearest science and says what is not shown: "compensatory renewal settles and strongly overcompensating renewal can oscillate (Ricker 1954; Beverton and Holt 1957; May 1976). These laws give the first's rest on a band and locate the second's turns; they do not show that overcompensation oscillates." (a result record)

How far seen, and use. Theorems; T1's law in the model (S-14). "The margin of a regime is harvesting under depensation and the fold (Clark 1976; May 1977), in whole numbers." (ch. 8) On data: "In assessed fish stocks the frozen-rest margin over the net draw ranks the risk of a break but does not time it" (ch. 2).

Known weaknesses.

- "How giving depends on the draw, or that giving is read off the level within a regime: both are premises, measured in yeast, not theorems." (a result record)
- "The fold over the reals, or the other order of a step (giving before the draw): neither is stated." (a result record)
- Of the margin read with the rest's exchanges held: "An identity can only be checked as accounting, not tested." (the experiment's README)
- Of the rises: "The superadditive rise of the point under a joint draw is the convexity of the inverse of a concave, increasing renewal." (a result record) MQ1's simulated Allee yeast measured harms, read by removal, not the point's rise: "The harm two harvesters did together exceeded the sum of their separate harms in all 12 cells (A4a–c)" (the experiment's README) (A4a–A4c passed, 12 of 12, 4 of 4 and 8 of 8). The theory's tests have measured neither in a real system.

Questions.

1. Does exactness in whole numbers, for any giving law, add anything to the continuous analysis of harvesting under depensation?
2. Is the draw taken before giving the right order of a step for the systems you know, and would the other order move the point of no return?
3. Has the superadditive rise of the point under added draws, which the record reads as the convexity of the inverse of a concave renewal, or the super-additivity of harms MQ1 measured, been measured in a real system?

S-13 The optimum laws for a renewable resource

```
theorem frontier_holds {better : (Nat → Nat) → (Nat → Nat) → Prop}
    (renews : ∀ z, R z ≤ r)
    (rising : ∀ a b, drawnOver R a (n + 1) < drawnOver R b (n + 1) → better b a)
    (greatest : R y = r) (reach : y ≤ s + R s) (drawsDown : s + R s ≤ y + k)
    (holds : y + r ≤ k) (long : 0 < n) {x : Nat → Nat}
    (front : Frontier (OpenCourse R k s) better x) :
    x (n + 1) = 0 ∧ ∀ t, t < n → R (x (t + 1)) = r
```

File: `Becoming/Optimum.lean`.

Meaning. (any `R`; the breaks on the World) Renewal is a law `R` of the level, at most `r` a step; `y` is a level of greatest renewal; `s` is the start, `k` the capacity, `n + 1` the steps of the horizon; `x t` is the level a course leaves at step `t`.

- *Over a horizon* (the block, `Optimum.frontier_holds`). Every optimum of every order rising with the total drawn holds a level of greatest renewal at every step before the last and leaves nothing at the end, given reach and `y + r ≤ k`. A need above every such level breaks at the first step (`Web.frontier_broken`).
- *Weighting later draws less* (`Optimum.frontier_exhausts`). With a capacity covering `s + R s` and `R 0`, where the weight `w` of a later step's draw falls, at every step, faster than any positive level can renew (`w (t + 1) · (z + R z) < w t · z`), every optimum of every order rising with the weighted total leaves nothing at every step from the first to the `n`-th. Conversely, where `R 0 = 0` and some level `y` within reach, with `y + R y ≤ k`, renews faster than the weighting falls (`w t · y < w (t + 1) · (y + R y)` at every step), every optimum leaves something at the first step (`Optimum.frontier_leaves`): "the discrete form of a discount rate above the greatest rate of renewal making extinction optimal" (ch. 5).
- *One step* (any set of outcomes). Every optimum of every order rising with a family stands at the least eligible level beyond, condition by condition, exactly where room holds (`Optimum.least_iff`): room is necessary as well as sufficient. Over one step where a part's members take at most their capacity and give no more than they take, the rest exchanging as it does, room holds, with equality of conditions decided and the family set apart from what lies beyond (`Optimum.Above.room`). The alignment dossier treats this law (A-02).
- *Several optimizers.* At a mutual best response a condition beyond them all stands as under one optimizer of their joint capacity (`Optimum.mutual_as_one`). Where others first draw from what an optimizer leaves, by a rule supplied, what they leave with its renewal being at most the level and a capacity covering what stands at the start and at nothing, every optimum of every order whose weighting falls at every step ends the stock at once, over a horizon of at least two steps (`Optimum.Next.frontier_ends_drawn_first`); in a witness, under weights `w (t + 1) ≤ w t < 2 w (t + 1)`, "a doubling source is rebuilt by every optimum alone and ended at once with others first drawing all but a quarter: the discrete form of open access dissipating what a sole owner would keep (Gordon 1954)" (ch. 5). Drawers in turn with no floor derive the case in which the others leave nothing: they draw everything at every perfect profile (`Optimum.InTurn.perfect_at_floors`).

How far seen, and use. Theorems; a third part (S-02). "over a horizon, far-sight keeps only the base at its greatest renewal; a coupling fixed in total is outweighed by a long enough horizon (known in bioeconomics)" (a study record). For one step: "The first is the drive above, at optima rather than limits of

paths, with shared units for room." (ch. 5) (the first being Zhuang and Hadfield-Menell's (2020) Theorem 1). Where each part is known, in the audit's list: "The maximum sustainable yield (Schaefer 1954); the sole owner (Scott 1955); open access (Gordon 1954); extinction under fast discounting (Clark 1973); the most rapid approach path (Spence and Starrett 1975); the dynamic commons (Levhari and Mirman 1980)" (a study record). What it counts as the theory's own: "every order rising with what couples in, not one objective; room necessary as well as sufficient on any set of outcomes; room derived from one step of exchange; and the break read as a law of the dependence web." (a study record)

Known weaknesses.

- "with a discount the optimum may hold below the levels of greatest renewal." (a result record)
- "a mutual best response (its existence is not shown in general)" (ch. 5).
- The audit's one unlocated candidate, a coupling for each step a dependence is met, given to several optimizers: "False as UE0 states it." (a study record) What survives is "a specialization of threshold provision with a pivotal contributor" (a study record). Since 3 November 2026, after edition 1, 05 states that coupling for several drawers (05): each counts, besides what it draws over the steps that remain, `c` for each step at which what all leave meets a positive need `m`, nothing valued after the horizon. In turn, with two drawers or more and no floor, where `c < m` every perfect profile leaves nothing at every step before the horizon (`Optimum.Coupled.drains_below`); under 05's gate (below the need what stands next is below it, at or above it what stands next meets it, and what stands next from nothing, with the need, is at most what stands next from the need), where `m < c` every perfect profile keeps the need at every step from a start meeting it (`Optimum.Coupled.kept_above`, `Optimum.Coupled.kept_levels`). On a gated renewal, renewing by `q` at or above the need, where `c < m` and `0 < c + q`, one drawer over two steps keeps the need at the first step where two drawers or more in turn drain it (`Optimum.Coupled.one_keeps_several_drain`); several optimizers draining as one of their joint capacity "is shown for one step and for courses re-optimizing at each step, in another model; this contrast is inside drawers in turn" (ch. 5). Drawing at once, where no portion reaches the need and none returns more than one for one, drawing everything is perfect at every `c`: "no drawer is pivotal (Palfrey and Rosenthal 1984)" (ch. 5) (`Optimum.Coupled.AtOnce.unpivotal_drain_perfect`). Not shown: "the keeping side without the gate, which profiles are perfect at once where the drain is not, and that the drain is the only perfect profile there" (ch. 5).
- [this dossier, AI] The horizon law assumes a capacity of at least `y + r` and an undiscounted total, and so gives the policy of greatest sustained yield with the stock drawn out at the horizon. Below that capacity it is not stated.
- [this dossier, AI] No cost of drawing is represented: the orders rank by the (weighted) total drawn. In bioeconomics the cost decides both results quoted. Clark's extinction under fast discounting needs, besides a discount rate above the greatest rate of renewal, a price above the cost of harvesting at low stock; Gordon's open access dissipates the rent, and with a positive cost it settles at a positive stock, the bionomic equilibrium, not at nothing. The two quoted readings are their limits at zero cost.

Questions.

1. Do these laws add anything to bioeconomics (Clark 1973; Clark 1976) beyond exactness for every renewal law and every order rising with the draw?
2. Is the condition of the weighted law the right discrete form of a discount rate above the greatest rate of renewal?

3. Which of them would a resource economist expect to fail for an actual fishery, and would a cost of drawing change them?

S-14 T1 and its tests

Meaning. Thesis T1, persistence certifies the joint draw: "Whatever persists through many renewal cycles of a condition it depends on, as many as each test fixes before it runs, renewal stopping below a need, has had the joint draw on it, its own with every other drawer's, kept within the regime's margin (its point of no return, not the withheld-levels margin)" (ch. 2). In the model this is S-12 with S-11. It forbids "a condition below a need measured in advance returning with nothing given from beyond; a dependent active while its condition stood below its need" (ch. 8).

How far seen, and use. Thesis, proved in the model and beyond it conjectured, with "no field test of a need measured in advance" (ch. 8), and "On yeast values not read before, no frozen test of the need or of the fates it separates has passed" (ch. 2). The yeast data are those of Dai et al. (2012) and Dai, Korolev and Gore (2015); the fish stocks are from the RAM Legacy database (tests). The tests' summary for the laboratory: "In the laboratory, it predicts thresholds, and the level is the state." (the experiment's README) Past the fold, "the populations broke three and at least six days before the renewal curve measured in constant regimes said." (the experiment's README) The audit: "The strongest empirical contact runs through T1." (a study record)

The frozen tests as they fell (ledger):

- MQ1, simulation: its two harvesters (A4), "each within the largest sustained draw and together beyond it, collapsed the population, as follows in the deterministic limit once their draws add, and their harms together exceeded the sum of their separate harms in every cell, which could have failed" (ch. 2). Its other passes bearing on T1 (A2, the break day within its band; A3, the point of no return) are deterministic-limit checks on a model built with a need, testing the measured quantities under noise, not the thesis.
- MQ1, real yeast: 1 of 2 passed. D2, the sharp need (above 1.5û alive, below û/1.5 extinct), failed: 19–46% of low starters read above zero, at the instrument's floor.
- MQ2, real yeast against held-out regimes: 4 of 8 passed, the need and its rise with the draw (M, A1, A2, B2). The levels failed (S and S2, 1.52 and 1.55 times off at DF 1600), and so did the fates after a salt shock (C1 and C2, accuracy 0.78 and 0.81, the null's 0.78). Its self-review: "What passed is the need. The level failed" (a study record).
- MQ3, the regime's step in 113 assessed fish stocks: 0 of 4 passed. F0 failed (last year's escapement added 0.22, against a bound of 0.10), and so did F1 and F2 (the one-year and five-year forecasts, each beaten by the frozen rest). F3 predicted that under a held catch at least 80% of episodes below the regime's point fall and at least 90% above it hold; of 754 episodes below it, 63% fell (99.5% of those above held).
- MQ6, yeast at constant regimes and past the fold: 3 of 10 passed. A1 (fates) and A2 (the held-out need) failed: the frozen fate reading "counted noise at the instrument's floor as survival, and A1 failed on it" (the experiment's README), and the held-out need came out "1.7 times too low at 0.2%, next to the fold (A2 failed)" (the experiment's README). Five of its six forecast and timing predictions failed (test labels): T1a-DF, T1b-DF and T1b-SUC, the regime margin's forecasts of the level away from and near the break against the frozen rest, persistence and trend, and both timings of the break, T2-DF

and T2-SUC. T1a-SUC (the forecast away from the break under falling sucrose), C1 (the regime's level carrying across experiments within 15%; 8%) and L (the level as the state at a constant dilution) passed.

- CG1, the gated commons: its passes bearing on T1 are of the identity class, which "gives no empirical support to the theory beyond the correctness of its identification" (the experiment's README), and the scope prediction A2, which passed (S-11).
- The fish-stock result the tests' summary gives, "In managed fish stocks, it ranks the risk of a break but does not time it." (the experiment's README), is the frozen-rest margin's (MQ4's prediction labelled T1, a test label), and that margin is the withheld-levels margin T1 excludes; a break there is falling below half the level within ten years (S-12).
- T1N and T1R, on the walleye lakes of Wisconsin's Ceded Territory: designed and frozen, and each ran once on 6 November 2026. "A field test with the need fitted on the years to 2012 and read on the years after, on the walleye lakes of Wisconsin's Ceded Territory, is designed and frozen, and its run refused at the header of the agency's public table of adult estimates, before reading a row: the table gives no upper bound and no acres column, and on it with the stocking database, as the frozen reader reads them, its rules could read at most 9 lakes below the need with a fate resolved, whatever the table's values, where the verdict needs 10" (ch. 2) (T1N, the record). "A second, on whether the lakes' natural renewal returns once stocking has raised their adults back above that need, is designed and frozen, and refused at the same point" (ch. 2) (T1R); "it shares lakes with the first, so the two are not independent results. Neither says anything of T1." (ch. 2) So no need was measured and no fate read, and no field test of a need measured in advance has run.

Known weaknesses.

- "Depensation is rarely detected in assessed fish stocks (Myers et al. 1995), so, at the levels studied, most show no need above the least that can renew at all. At that need the return T1 forbids is one from less than can renew, which cannot happen, by what that need is, where the level counts all that can renew: that item tests nothing on those stocks, and their showing no higher need does not count against T1. A dependent's own need, and the point under a held catch (§272, above), can still test them. The need is measured apart from the fate it predicts, so the scope is not chosen afterwards." (ch. 2) (§272 being MQ3.)
- "The held-out dilution factors had been read in MQ1, so they were held out from the fit, not from the analyst." (the experiment's README)
- "The paper was read before the freeze, its figures and tipping days included, so the timing verdicts were partly anticipated, and the freeze says so." (the experiment's README)
- Both fate failures at the instrument's floor (MQ1's D2 and MQ6's A1) failed under their frozen rules, and the account keeps them as failures. Read by the floor afterwards, A1 separates 97–100%, a post hoc reading; of every post hoc analysis, "None of them changes a verdict." (the experiment's README) The account's lesson from the second: "The lesson is MQ1's again: read the instrument's floor before freezing a fate rule." (the experiment's README)
- The salt-shock test (MQ2's C1 and C2): fates after the shock were predicted with accuracy 0.78 and 0.81, the null's 0.78, and T1's refutation clause names "fates across a predicted point separated no better than by a null" (ch. 2). After the shock, "on the first reading, every population stood above the predicted need, so the need called fates as the null did (0.78), meeting the letter of *Refuted by*; with

none below the need the test could not reach what T1 forbids, and each error was a death above the need" (ch. 2). The chapter reads the result as "its first prediction: 7 of 8 died; the converse, not claimed: a shock changes the regime and its need" (ch. 2). That change of regime rests on a post hoc analysis, "the salt-shock experiment's pre-shock levels are 8-20% below the bifurcation experiment's" (a study record), with needs located for each column by its own level right at DF 825 (7 of 8) and wrong at 750 (3 of 8). And "Four days after the shock is too short to settle fates near the need." (the experiment's README)

Questions.

1. What field test of a need measured in advance would be feasible, and in which system? (One designed on Wisconsin's walleye lakes could not read the public table: the last of the frozen tests above. On 30 of T1N's 48 simulated worlds, put on the records' grid at its widest, with GLIFWC's estimates added to the agency's table (V1 being T1N's verdict), "So a successor keeping T1N's rules by hash would, in those worlds, reach a verdict on V1 once in 1,200 archives, and never in T1's own world." (the experiment's code) "Whether the lakes would is not known" (the experiment's code).)
2. Does T1 forbid anything an Allee-effect model of the same system does not?
3. Was the salt-shock test (Known weaknesses) a fair test of T1, and how should its result count?

S-15 T2 and its tests

Meaning. "T2. The state of a renewable condition under a draw is its level, its recent giving and one lag, its place named before the data" (ch. 8). It forbids "a fourth variable from the level's own history removing a tenth of the residual variance in the median series; the level alone forecasting as well as the three numbers at five steps of the grain at which they are read, fixed before the data (five years in the fisheries), in a series not shown to sit at the regime's-step end (where, by a test fixed before the data, the upper confidence bound on what the recent giving and the lag together remove, beyond chance, of the residual variance the level leaves in the next giving is below a tenth); the lag's coefficients changing sign between series" (ch. 8). "Beyond delay-difference models (Deriso 1980; Schnute 1985) and partial adjustment, it claims these three numbers for every renewable condition under a draw; silent on coupling, which adds a partner's level." (ch. 2)

How far seen, and use. Thesis, conjectured. The frozen counts (ledger): MQ4 passed 7 of 12, failing G3 (its whole-state verdict: nothing needed beyond the level and the recent giving), F5a and F10a (the five- and ten-year directions), F10b (the ten-year error) and T2 (a test label: the frozen-rest time to break, half of the origins put within five years breaking within ten; 0.32); MQ5 passed 6 of 8, failing D1 and D2 (the draw's response); MQ7 passed 0 of 1: C1 failed (0.39, lower bound 0.00, on 62 episodes from 5 stocks), and C2 was not testable. UE32 "found T2's numbers better than the field's fitted embedding by its frozen criterion, both behind persistence in most stocks, the embedding losing more with longer history: sufficiency against it at these lengths, not the lag's place or T2's law" (ch. 8) ("better in 42 of 65 stocks, $p = 0.025$; expected: not separated; post hoc, within the field's bound on its size short of 0.05: 16 of 22 at $D = 2$, $p = 0.052$" (ch. 8)), the size being the embedding's. UE31b's verdicts on the flour beetle's cultures are in the first weakness below. In yeast both chapters place MQ6's constant dilution outside the settings T2's *Refuted by* names: "in MQ6's yeast, residuals at one constant dilution, never past the fold, carry nothing day to day, outside the settings T2's *Refuted by* names, and the runs past the fold measured none of its forbidden items" (ch. 8); the home chapter gives the measure, "residuals at one constant dilution, never

past the fold, carry nothing day to day (lag-1 −0.14, passed)" (ch. 2). Of MQ5's pass: "the lag may be the assessment's own, the gain damping" (ch. 2). "no Lean statement excludes what T2 forbids, so it is a conjecture, not a third part" (a study record). Restated on 4 November 2026: "No completed verdict changes" (ch. 2). Narrowed on 6 November 2026, on UE31b's failure, the first forbidden item having read "a fourth variable removing a tenth of the residual variance in the median series": "That forbids less: a measured variable outside that history, such as a stage before recruitment, may now carry more without counting against T2. The failure stays on record as a loss of the broader claim, not a narrowing chosen in advance. A delay embedding of the escapement and the giving is more of the level's own history, so one that forecast better would still count against T2. No completed verdict changes." (ch. 2)

Known weaknesses.

- T2 first claimed that the three numbers, with the draw, fixed the next giving "as far as anything measured does". That wording failed on a frozen run (UE31b, on 6 November 2026) in one domain, the flour beetle's laboratory cultures, where a stage before recruitment is counted, since, "the pupae counted now are the next recruits and carry the chance events between egg and pupa, which no history of the adults can hold, so a class of fewer variables, the pupae and the adults, left less of the next giving unaccounted for" (ch. 2). "The beetle is not on the *Refuted by* list, but a culture under a set loss is a renewable condition under a draw, so the failure counts against the claim for every one; it defeats T2's class as worded in that domain, not the reading 08 §9 names, nor T2 in the fisheries, where MQ5's verdicts on the state and the lag passed (its two on the draw's response failed)." (ch. 2) That reading is "reading a subject as continuation relative to a class explains how it goes on" (ch. 8). UE31's design says what the field's state is there, "The field's state there is the beetle's life stages (the LPA model, §3)" (the experiment's design). UE31's own run, on 3 November 2026, decided nothing, its adult-loss cultures being too short for its rules; UE31b read the same cultures by cell (result). "Its reading is the one it expected" (ch. 2) (the rival being the field's state, the pupae and the adults), "separated on the next step: the rival's state; three numbers suffice on the course" (ch. 2). "Beyond the three numbers, the pupae and the adults removed 0.83 of the residual variance beyond chance (the median across the five set losses, from 0.36 at the lowest to 0.91 at the highest, detected in 8 of the 10 cells); beyond the pupae and the adults, the three numbers removed 0.004 (detected in none); out of sample the field's state had 0.06 of their squared error in the median cell, and less in all 10 (reported). A difference between replicates that neither law carries moves this toward the field's state, since four replicates share one law a cell (UE31b §3, §8)." (ch. 2) Of that run's limits, "the rival's constants come from other analyses, the unit is four replicates under one law, and the model that wrote the design may have met these cultures' series in its training text" (the experiment's README). What remains is that, with the draw, the three numbers fix the next giving "as far as the level's own history does" (ch. 2): "on the adults' own course, with the lag two censuses back, where the life cycle puts it, a further lag removed 0.05 against the bound of a tenth (detected in one cell; 0.18 at a set loss of 0.27), so the three numbers suffice there." (ch. 2) "That is a prediction that could fail and did not, but there the three numbers are the field's own state seen from the course, a tie by construction that the field's model predicts the same way, so it shows nothing the field does not, and it is not T2 passed: of T2's own items the lag's change of sign was not testable (detected in 3 cells, 6 needed) and the level alone at five steps was not decided (the three numbers better in 5 of the 10; median error 0.27, the level alone's 0.23), and at this shape the sign could not go to the field's model, nor V4 fail T2 in the field's simulated worlds (UE31b §5, §6)." (ch. 2) (V4 is the level alone at five steps; the sections are UE31b's

design's.) "Their one contrary prediction, the lag's sign, was not decided by this archive (the two rules differ in 4 cells, 6 needed), which says nothing for T2's sign." (ch. 2)

- The evidence on a wild condition is fisheries alone. How far it has been seen, in 08's words: "conjectured, on four frozen fish-stock tests; untested on a wild condition outside fisheries" (ch. 8).
- [this dossier, AI] At constant dilution the level alone was the whole state: the regime's-step end of T2's law, at which recent giving and the lag carry nothing. The result is consistent with T2's law, and what it measured, the residuals' carry from day to day, is none of the three items T2 forbids as worded. T2's refutation clause names yeast only in deteriorating environments, "any of these on survey series of fish stocks, grazed grassland, harvested forests, or yeast in deteriorating environments" (ch. 2), but "it claims these three numbers for every renewable condition under a draw" (ch. 2), and a culture at a constant dilution is one, so a forbidden item measured there would count against it, as T2's sentence on the flour beetle reasons of a culture under a set loss. Since the level alone was the state there, the second forbidden item, the level alone forecasting as well as the three numbers at five steps of the culture's grain, could be met in such a culture. The chapters now lift that item where a test fixed before the data shows, by a bound, a series at the regime's-step end; no such bound was computed at MQ6's constant dilution (question 4).
- "The series are assessment outputs, not measurements; age-structured assessments carry a recruitment lag of their own." (the experiment's README)
- Of MQ7: "So the field frequency of fates on either side of the curve is still not measured." (the experiment's README)

Questions.

1. Is the one-lag state a property of the stocks, or of the age-structured assessments that produced the series?
2. Which fourth variable from the level's own history would you expect to remove a tenth of the residual variance first?
3. Which survey, grassland or forest series would test T2 fairly?
4. T2's law has two ends, "the regime's step and the frozen rest its two ends" (ch. 2), and this dossier reads MQ6's constant dilution as the first, where the level alone is the state. The second forbidden item now excludes a series shown to sit there, by a test fixed before the data (Meaning). Is that the right mark, given that a series too short or noisy for the bound stays within the item, so that T2 is most exposed where the data are poorest?

S-16 T6: a break seen coming

```
theorem break_not_bounded [DecidableEq C] {c : C} (unobserved : ¬ obs g c) (u : Nat)
    (others : ∀ s, s ≤ u → ∀ e, e ≠ c → W.requires s g e → W.need s g e ≤ W.level s e) :
    ∃ d, (∀ s, s ≤ u → (W.raise c d).Supported s g) ∧
      ∀ t, W.Supported t g → W.AlikeThrough obs g W.level (W.raise c d).level t
```

File: `Becoming/Runaway.lean`.

Meaning. "T6. A break is seen coming only through what couples into what breaks" (ch. 8). "A statistical early warning reads the observed condition's own rate of return, which falls near a bifurcation of its own

renewal; a dependent's need is no such point, and before a dependent's break the stock's statistics rise as before the same drawdown with no break." (ch. 3)

- *The invisible half* (World; the block). For a condition `c` the formation `g` does not observe, its other requirements met up to step `u`, some raise `d` of `c` keeps `g` supported to `u`, while the raised and unraised histories stay alike through `g`'s couplings at every step at which `g` is supported (`Web.break_not_bounded`).
- *The seen half* (periphery, `warning/SlowingDown.lean`). A linearized deviation stepping by a multiplier `a` plus noise, deviation and noise independent with finite second moments, has, where its variance is the same across the step and `a² < 1`, stationary variance `Var E/(1 − a²)`, and lag-1 autocorrelation `a` where that variance is positive (the measure finite); at a fold, or at a transcritical end under a proportional draw, `a` rises to one as the margin to the condition's own end shrinks. "A dependent's need enters neither statistic, by construction: no one function of the multiplier returns the margin to a need for every need." (ch. 3) With the model known the multiplier does give it: in the logistic drawn model with h, r and K nonzero, a measured need's margin is read off the multiplier (`SlowingDown.need_margin_read`).

It forbids "Given the stock's own course, its early-warning statistics separating runs in which a dependent's need is crossed from runs in which it is not, the dependent taking no measurable share." (ch. 3)

How far seen, and use. Thesis; the invisible half proved on the World, the seen half in the periphery, in the linearized and logistic models only; that the split is one law across domains, and that a break is seen coming only so outside these models, are conjectured (08 §9, 03). "EW1, the literature, consistent; three frozen seabird tests did not reject it at their thresholds: EW2 at low power, estimates in the refuting direction; EW3 near zero, not independent of EW2 (it reuses part of its rows); EW4 at moderate power; none establishes conditional independence" (ch. 8). The frozen counts (account): EW2, EW3 and EW4 each passed both of their verdicts, V1 (the lag-1 autocorrelation of the prey's window) and V2 (its variance). The dependent's need: "Cury et al. (2011) measured such a need: seabird breeding success falls below about a third of the largest forage-fish biomass." (the experiment's README) The chapter's credit: "Critical slowing down (Wissel 1984; Scheffer and others 2009); nonlinear runs, finite windows and several variables are not treated." (ch. 3) The closest prior work, in the early-warning check: of Dekker, von der Heydt and Dijkstra (2018), in whose one-way coupled bistable systems the leading system's indicators did not differ whether or not the follower tipped, "The closest prior form of the uncoupled side, with a matched comparison." (a study record) Of Boerlijst, Oudman and de Roos (2013), "The break is seen only in what carries the critical mode, not in what breaks: T6's coupled side in eigenvector form" (a study record). Of the first wording, "its first wording fails against unmatched controls wherever the stock's rate of return falls as it is drawn down (Kefi et al. 2013; Drake and Griffen 2010; Clements et al. 2017)" (a study record).

Known weaknesses.

- The literature refuted T6's first wording: "The literature (Dai and others 2012, 2015; Boerlijst and others 2013; Dekker and others 2018) is consistent and refuted the first wording." (ch. 3) The method chapter lists among prose that said more than the theorems "that nothing a part senses shows how near a break is. The theorem assumes the stock is not sensed, and the early-warning literature (Scheffer and others 2009) shows the general claim false" (ch. 8).

- "The pass is weak. Every estimate lies in the direction the refutation clause names." (the experiment's README) EW2's frozen p was 0.048 against a familywise 0.025: "AR1's one-sided p would have been below 0.05 as a single test; the frozen familywise rule set 0.025 for each of two." (the experiment's README) Reported only: "The combined partial correlation is 0.30 for autocorrelation (95% interval −0.06 to 0.59) and 0.26 for variance (−0.10 to 0.57), computed after the run from the frozen statistic" (the experiment's README). Of EW3: "EW2's positive estimate did not hold up at about twice the power; by the frozen reading rule the variance resolves it toward T6 and the autocorrelation leaves it open." (the experiment's README) Also reported only, and for one stock in the refuting direction: "On the rows EW2 did not test (R5), Benguela's estimates are 0.64 and 0.61 on 15 rows with six covariates, Røst's −0.15 and −0.30." (the experiment's README)
- Of EW4: "The scope premise is argued by mechanism, not measured." (the experiment's README)
- "no: the uncoupled side holds by construction in matched form and is the eigenvector literature's" (a study record).
- [this dossier, AI] Given the stock's course and no feedback from the dependent, the forbidden separation is excluded by construction. The empirical content lies in the premise of negligible feedback, which no test has measured.

Questions.

1. What does T6 claim beyond the case without feedback, and could data show it?
2. Are the three seabird tests fair, given their power and the reuse of rows?
3. Which system with a measured, negligible feedback from the dependent would test T6 well?

Part 5. Mind and inference

S-17 The free-energy principle as a case

Meaning. The free-energy principle in the theory places each of the principle's quantities and claims with its hypotheses and home section; its brief takes about eight minutes. In sum:

- *Stated.* Belief as conditioning; variational free energy, which "bounds the record's surprisal, with equality exactly at the posterior" (FREE-ENERGY-PRINCIPLE), in the periphery over the reals and, for beliefs in counts, in the pure core; expected free energy and its terms; the blanket as a coupling structure on a determined law, and its statistical form under sufficient conditions.
- *Forbidden.* "Five statements of the principle's literature, read as guarantees or unqualified, fail on models in the principle's own terms, each by a law the principle does not state" (FREE-ENERGY-PRINCIPLE): on identical cells, on one generative model with alike preferences, on risk minimization keeping a need, on seeking information correcting itself, and on agents coming to share a code, the third and fourth weakly.
- *Forbidding nothing.* "The existence claim forbids nothing alone, as its proponents concede" (FREE-ENERGY-PRINCIPLE). The theory's own root laws "hold of every carrier, relation and class, so they forbid no relation" (FREE-ENERGY-PRINCIPLE).

- *Not stated.* Densities, continuous time, Langevin flows, generalized coordinates, precision, hierarchical and deep temporal models and structure learning, among others: "none is stated in the theory." (FREE-ENERGY-PRINCIPLE)
- *What excludes a measurement.* Thesis T6 (S-16); the page's §6 says how a measurement reaches each kind of claim.

The rest of this item takes one law of the page, the first study's, as the case with its Lean: the habit. (World for the selection laws, counts for the count laws; in `Becoming/ChoiceCountCapture.lean`) A study (UE1) read the free-energy principle against the selection laws, on a simplified chooser (two policies, choice at the argmax, counts of the executed policy as its prior), with no probability; Known weaknesses give how active inference's own scheme differs. A choice made from any fold of what has coupled into a formation is a selection through its couplings (`ChoiceCountCapture.selectsThrough`); choosers sharing their folds select through their pooled couplings (`ChoiceCountCapture.pooled_selectsThrough`). On counts: a chooser over two policies takes the first where its rule weighs it at least as much as the second, the rule read from the counts of past choices weighted by a standing score, `κc` for the first policy and `κo` for the second. Under a standing score, with the rule in force read from the counts at the start, from counts at or above the threshold `κo · other ≤ κc · carrier`, along every schedule of reading, the course is locked at every step and neither the share nor the rule falls, even where the score favours the second policy (`ChoiceCountCapture.counted_locks`); below it the second policy is taken at every step (`ChoiceCountCapture.counted_locks_down`). Counted instead from the posterior at the time of choosing, a contrast case, a score favouring the second policy lowers the first policy's share strictly at each step at which it has carriers (`ChoiceCountCapture.posterior_count_falls`), and at every step of a course at a positive fixed scale (`ChoiceCountCapture.posterior_course_falls`). Joined to a coordinate that the loop does not read and that rises by at least one at every step, the course stays locked while the coordinate passes every bound (`ChoiceCountCapture.counted_outlives_loss`).

How far seen, and use. Of the page's laws: "Every law named is near-definitional, a known result stated in the principle's terms, a corollary, or a witness." (FREE-ENERGY-PRINCIPLE) And "In the third parts the theory reaches farther than the principle alone, in what it forbids; it states less than the principle where it states nothing yet, in densities, continuous time and deep models among others, and it does not yet reach farther than the sciences taken together" (FREE-ENERGY-PRINCIPLE). Of the habit, of the selection laws: "This part is near-definitional." (ch. 2) The lock "is the lock-in of increasing returns (Arthur 1989) and, counting the action taken, the value-free habit (Miller, Shenhav and Ludvig 2019); active inference's own simulation of devaluation shows the persistence (Friston and others 2016, §5.2)" (ch. 8). The verdict: "What the law adds is exactness for a simplified model and its edges, so as a containment of the free-energy principle it is a restatement" (ch. 8); the containment's third parts lie in 05 §9, 05 §1 and 03 §6.

The first study's search, before the page. A search of 16 claims in 15 papers found that "UE1's own laws forbid nothing found." (a study record) "Every collective claim found stays within the union of the members' couplings." (a study note) Its strongest candidate is the claim of Friston, Levin, Sengupta and Pezzulo (2015) that identical cells are guaranteed to reach unique places: "The commuting law forbids the guarantee for the paper's deterministic descent from an exchange-fixed start: the cells never part." (a study record) But "The paper's own simulation broke the symmetry with small random initial expectations." (a study record) Under noise of one law at every cell the law gives way to an exchangeable law, which decides

nothing of which cell takes which place and forbids nothing of the paper, whose scheme has that form (record). The lock's use for alignment is in A-07 and A-08.

Known weaknesses.

- Of the five forbidden statements: "Each restates a known result of a neighbouring field in the principle's terms." (FREE-ENERGY-PRINCIPLE) And "Each excludes a guarantee or an unqualified statement on a model, not yet a measurement of an actual system" (FREE-ENERGY-PRINCIPLE)
- "That the principle's partition is a coupling structure of the World model is a reading. No representation theorem joins a Langevin system with a blanket to a web." (a study note) And "Its proponents present the principle as a method of description rather than a claim that could be refuted, and its process theories as the part that makes predictions." (a study note)
- "That actual agents, brains or trained systems count their choices so or choose at the argmax; the overtraining effect the lock matches in rats has failed to replicate in humans (de Wit and others 2018)." (a result record)
- "The in-sync start is a hypothesis: counts above the threshold under a rule left from an older score lock the other way (`Witness.Stale.above_but_stale_locks_down`)" (a study note): three choices to one, above the threshold of the score one to two, under a rule an older score left in force, lock the other way from step 5.
- The chooser is not active inference's: "Active inference's own scheme: it counts the end-of-trial posterior over policies, close to the executed policy where outcomes reveal it and split among policies that explain the trial equally; its prior is the counts' expected log, not the counts; it has a free-energy term, sampling and inferred precision. The threshold is the model's, not SPM's." (a result record) Nor does it read "Any probability, surprisal, free energy or expected free energy" (a result record). The study adds "Policies that explain the trial equally share it, which the kit does not model; the paper places the habit's advantage exactly there." (a study note) Of the simulation the verdict cites: "the simulation shows persistence over the trials it runs, which is consistent with a lock and with slow decay alike." (a study note)
- The first study's grading, before the page: seven of the claims are forbidden under a natural reading by other laws of the theory (symmetry, joint excess, lumping). [this dossier, AI] These are the page's five: of the three statements on a shared language two make its fifth entry, and the third, the duet, is not counted. Of these, "Each forbids a verbal generalization, not a paper's simulation" (a study note). And "Each forbidding law is a known result, symmetry breaking (Curie, Turing, Angluin) or the commons (Lloyd, Hardin)." (a study note) The page also credits pivotality, identification and conventions symmetry does not select.
- The study reports the principle's critics (Biehl, Pollock and Kanai 2021; Aguilera, Millidge, Tschantz and Buckley 2022; Bruineberg, Dołęga, Dewhurst and Baltieri 2022), among them "Sparse coupling does not in general give the blanket condition." (a study note) The record: "the selection laws read its sparse coupling, not its blanket condition" (a result record). [this dossier, AI] So the selection laws read the principle's partition only as sparse coupling. The statistical form is stated apart, under sufficient conditions in counts and in the linear case. The step the critics dispute from a blanket to beliefs is not part of the containment.

Questions.

1. Is reading the principle's partition into thing and world as a coupling structure fair, given that the selection laws read sparse coupling and the statistical form is stated only under sufficient conditions?
2. Active inference counts the end-of-trial posterior over policies and reads its prior as the counts' expected log (Known weaknesses). Does its habit learning lock as a rule read monotonically from the counts of its choices does, or does the split among policies that explain a trial equally change that?
3. Is there a claim in the principle's literature that a proponent would accept as refutable, so that a third part bears on what proponents claim and not on a reading?
4. The page reads five statements as guarantees or unqualified. Would their authors accept any of these readings as what they claim? Where they would not, does anything stay forbidden?

S-18 Conditioning by restriction and the exit from a margin

```
theorem fallsWithin_share (p q m n : Nat) :
    (share (Typicality.courses n) (weight p q) (fallsWithin p q m)).le (n * (p * q), m * m)
```

File: `Becoming/ConditionalWeights.lean`.

Meaning. (any type; natural weightings of listed courses)

- *Conditioning by restriction.* The weighting restricted to a record keeps each course agreeing with it and gives every other nothing. Bayes's rule holds as a ratio identity (`ConditionalWeights.restrict_two_ways`); in a generative model a hypothesis's share given the record is its prior weight times its likelihood weight over their sum (`ConditionalWeights.posterior_single`); a prior weight is the prior share only where every hypothesis's likelihood weights have one total (`ConditionalWeights.prior_share`, `ConditionalWeights.Witness.prior_not_share`).
- *The exit from a margin* (the block). A level stands `m` above a need on courses of `n` independent steps, raised `q` by each one and lowered `p` by each zero, so that rises and falls balance under the weights `p` and `q`. The courses on which it reaches the need take at most `n·p·q/m²` of the weight (a share is compared with a pair by cross-multiplication): "Kolmogorov's maximal inequality restated in counting form" (ch. 2). Given a record of the earliest steps that leaves the level `m′` above the need without having reached it, the share of the courses reaching the need within the `n` steps left is the share from `m′` (`ConditionalWeights.record_share`).

How far seen, and use. Theorems. "These are identities of counting" (ch. 3). "Bayes's theorem in counting form (Bayes 1763; Laplace 1774) holds as a ratio identity, near-definitional" (ch. 8). "The exit bound is the unregulated case, not the free-energy principle's quantity, and gives that principle's containment no third part." (ch. 8) For T6 (S-16), "the weighting is the independently warranted model T6 allows" (ch. 3).

Known weaknesses.

- "Averages of logarithms, entropy and free energy among them, are not the logarithm of one reciprocal share" (ch. 3); for beliefs in counts the free-energy bound has a counting form, with its equality, in the pure core, since 30 October 2026.
- "That these identities are substantive results of a containment of the free-energy principle: they are identities of counting, near-definitional." (a result record)

- [this dossier, AI] Natural-number weights confine the laws to rational probabilities over finitely many listed courses.
- [this dossier, AI] Kolmogorov's inequality is the weakest of the standard maximal bounds: for steps bounded as here, exponential martingale bounds (the Azuma–Hoeffding inequality) are much tighter in the tail. For one of the audit's candidates, survival under expected joint excess, UE6's first round found a sharper statement known: "Bertsimas and Popescu 2005 (sharp one-sided moment bounds, stronger than the Chebyshev form)" (a study record).

Questions.

1. Does a counting form of Bayes's rule and of the maximal inequality give inference anything beyond a constructive proof?
2. Averages of logarithms are not the logarithm of one reciprocal share, while for beliefs in counts the free-energy bound has a counting form. Is the first decisive for containing the free-energy principle?

S-19 Orders with a set point

Meaning. (any set of outcomes; the draw on the World; in `Becoming/SetPointOptimum.lean`) An order is *nearing* a set point `s` of a family `ds` where it ranks above every nearer outcome, condition by condition (`Optimum.Nearing`). Below the set point a nearing order drains as a rising order does: among outcomes up to the set point its optima stand at the least level beyond exactly where room holds there (`Optimum.least_below_iff`). Where some eligible outcome reaches the set point, an outcome is an optimum of some nearing order exactly where it is eligible and at the set point (`Optimum.optima_iff`). In a balanced step in which the whole gains nothing, what lies beyond then falls by the family's shortfall from its set point with what the step loses (`Flow.Balances.atSetPoint_draws`). Formations holding a level drain a commons gated at `m` wherever their losses together exceed what the rest gives it at each of `n` steps from a level below `n + m` (`Flow.held_excess_ends`).

How far seen, and use. Theorems. The chapter marks `Optimum.optima_iff` near-definitional: "the class asks nothing of what lies beyond" (ch. 5). "What protects what lies beyond at the set point is the choice among equal optima, which no law here models." (ch. 5) "the plateau decides, not the set point." (ch. 5) Neighbours: such orders "have a plateau where the set point is reached, as an aspiration level has (satisficing; Simon 1955)" (ch. 8); "that what protects at the set point is the choice among equal optima is mild optimization's point (quantilizers; Taylor 2016), its bound a reading not proved here" (ch. 8); "The drive of homeostatic reinforcement learning, a norm of the deviations from set points (Keramati and Gutkin 2014), is a nearing order." (ch. 8) "Neighbours, not containments: the laws restate them for one step, exactly, with the drain below the set point and the draw at it." (ch. 8) UE6's first round found the joint-excess side stated qualitatively within active inference: "Albarracin and others 2024 (component resilience depleting a collective's resources under active inference)" (a study record). A reading for active inference [AI reading, D45]: "Such a chooser escapes the optimizer's drive at its set point, not the drain below it, nor joint excess." (ch. 5)

Known weaknesses.

- "It does not show which optimum is taken, that any agent's preferences form a nearing order, or anything over a horizon or under noise" (ch. 5).

- "That a set point protects what lies beyond: in a witness where a step can lose what it takes, the order by distance has the draining step among its optima beside the keeping one; what protects is the choice among equal optima." (a result record)
- [this dossier, AI] Active inference ranks policies by expected free energy, which adds an epistemic term to the term of prior preferences; the reading takes the preferences alone.

Questions.

1. Are active inference's prior preferences over outcomes, and the drive of homeostatic reinforcement learning, nearing orders in this sense?
2. Is a law of one step of use for agents acting over a horizon under noise?

S-20 Selection through couplings across fields

Meaning. (World) With no premise, "Two histories are selected alike by every selection through a formation's couplings exactly when they are alike through them" (ch. 3) (`Web.selects_alike_iff`). With a part withheld and the rest of the exchange fixed, balance at each step on the conditions coupled into members of a formation's coupling closure, and every member of the part beyond that closure, every selection through the couplings of a formation within the closure treats the levels with and without the part alike (`Flow.unselected_beyond_coupling`); where the rest responds instead, the same holds under further hypotheses (`Flow.unselected_beyond_coupling_responsive`). Over cohorts of descent, "Counts of offspring see only what the selection couples into" (ch. 4), and where the whole gains nothing across the step and more offspring go only to formations raising a family, the share reaching beyond it is never selected against (`Flow.share_not_against`). The first unification: "reward, training, natural selection, prices and votes as one selection through couplings; the drain lies wherever room lies (the commons in evolutionary biology, Rankin, Bargum and Kokko 2007)" (ch. 8).

How far seen, and use. Theorems; across domains, "the laws proved; one law across domains conjectured (T3)" (ch. 8). "The main law is the proxy problem ... It holds by construction; its force lies in the supplied premise that an optimizer selects through just those couplings, and in what couples in." (ch. 3) Credited to "supervisory control (Ramadge and Wonham 1987), under partial observation (Lin and Wonham 1988; Cieslak, Desclaux, Fawaz and Varaiya 1988)" (ch. 8). Its use for alignment: A-01, A-02.

Known weaknesses.

- "the selection half is observability restated; the drain is row 20's third part" (a study record).
- "A particular selection may ignore what its couplings carry." (ch. 3)
- [this dossier, AI] For natural selection what couples in is whatever affects reproduction, so the law says that selection does not act on what does not affect reproduction.

Questions.

1. Does the law say more about natural selection than Price's covariance already says?
2. For control theory, is it observability under partial observation with nothing added?
3. Is one selection across reward, training, natural selection, prices and votes a useful unification, or does it erase what tells them apart?

Part 6. Order and open-endedness

S-21 Eventual widening

```
def EventuallyWidens (l l' : Nat → C → Nat) : Prop :=
  ∃ T, ∀ S, T ≤ S → (W.after S).Widens (tailFrom S l) (tailFrom S l')

theorem eventually_neither (widens : W.EventuallyWidens m l') :
    ¬ Verifiable (fun r : Run (Sequences.process (C → Nat)) => W.EventuallyWidens r.act l') ∧
    ¬ Refutable (fun r : Run (Sequences.process (C → Nat)) => W.EventuallyWidens r.act l')
```

Files: `Becoming/EventualWidening.lean`, `Becoming/EventualWideningInstances.lean`.

Meaning. (World; witnesses) One level path eventually widens another where, from some step on, it widens it on every tail (the definition): "first-order widening read on late tails, not a comparison of growth" (ch. 7). The theorem: where some course eventually widens a second, eventually widening it is neither verifiable nor refutable from finite records of a course. A course whose admissions a list holds eventually widens nothing (`Web.not_eventually_of_listed`). A course keeps widening where nothing admitted is lost and something keeps being opened (`Web.KeepsWidening`); over a listed type of formations, no course read by name does (`Web.KeepsWidening.not_listed`). Of the lag, the theorem: "a course keeping every formation of another from one step later is mixed with it in every tail (`LagWitness.lag`)" (ch. 7), so eventual widening is silent on it. That eventual widening needs no growth, a steady course eventually widening one that collapses, follows from the definition and is not a Lean statement; 07 gives it in a reading [AI reading, D45].

How far seen, and use. Theorems. "Every law here is near-definitional, and ranking eventual widening above is a requirement beyond C" (ch. 7). The method chapter: "is the inclusion form of a tail criterion, and a criterion reading only tails is a dictatorship of the future (Chichilnisky 1996)" (ch. 8). "That its verdicts are dense both ways is liveness in Alpern and Schneider's sense (1985)." (ch. 8) "That, with finitely many admissions at each step, it entails overtaking (von Weizsäcker 1965; Gale 1967) and not conversely is an argument, not proved in Lean." (ch. 8) And "Every law of the record is near-definitional; what it adds is its coherence with widening and its witnesses, the lag among them." (ch. 8) Of what open-endedness research calls innovation (07 §2): "That the possible with what is held grows only across a change of what is held is open-ended novelty's innovation, novelty that changes the model and with it the possibilities of variation (Banzhaf and others 2016), which holds there by definition" (ch. 8).

Known weaknesses.

- Of the second-order direction of widening that 07 has asked about, a range whose widening keeps widening: "Eventual widening, first-order widening read on late tails, does not give it." (ch. 7) On the ranges courses admit (record), "ranking every course that keeps widening by name above every course that admits alike, was wider at first and is mixed with it is not a strict partial order containing widening on every web: on a web of formations on two sides, whose static course admits infinitely many formations, it closes a cycle, and where the static course's admissions are listed it is one on every web." (ch. 7) Of the lag: "Eventual widening's lag is decided by widening with a lag and by a tail order containing eventual widening; on the own-condition web widening with a lag and eventual widening hold of a pair in opposite directions, so the lag does not choose between counting early steps and reading only tails." (ch. 7) Of ranges compared by what a course keeps reachable (07 §2): "covering,

the comparison of what courses keep reachable at the same offsets, implies inclusion of what each configuration admits now, so widening what can be reached never puts a configuration above one that starts a course and admits now what it does not" (ch. 7), and "read with a lag it decides one witness's pair, though not every such pair, and C asks nothing of that pair, so the lag's verdict is a requirement beyond C" (ch. 7). On causal-set growth "every course keeps widening what can be born next, so the second order is not told apart there" (ch. 7). "The candidate left is what carries a course's openings." (ch. 7) What one course keeping widening needs is S-22.

- Its record, among what it does not show: "That eventual widening is better: ranking it higher is a requirement beyond C (C∞ in the study), named, and no result here takes it" (a result record).
- [this dossier, AI] A course admits a formation where its levels meet every requirement the formation has (`Web.Admits`), so the laws concern which formations a course's levels could support, not which arise; measures of open-endedness in artificial life usually count what arises and persists.

Questions.

1. Is admitting more formations at arbitrarily late steps a useful notion of open-endedness, compared with the measures used in artificial life?
2. Verdicts neither verifiable nor refutable from a finite record make eventual widening a liveness property. Do the definitions of open-endedness used in artificial life share this, and is it discussed there?
3. What measurable criterion would capture a range whose widening keeps widening?

S-22 What keeping widening needs

```
theorem KeepsWidening.passes_every_bound {T : Nat} (keeps : W.KeepsWidening itself W.level T)
    (req : ∀ t f, T ≤ t → W.Supported t f → W.RequiresWithin cs f) (K M : Nat) :
    ∃ t, M ≤ t ∧ K < W.stock cs t

theorem not_keepsWidening_along_passings (balances : L.Balances W) (acting : L.Acting W)
    (dependent : W.Dependent) (unloosened : ∀ t f, W.Unloosened t f) {R : Nat → F → F → Prop}
    (contains : ∀ t g f, Web.passingContinuation (L.PassesSupport W) t g f → R t g f) (T : Nat)
:
    ¬ W.KeepsWidening R W.level T
```

Files: `Becoming/KeepingWidening.lean`.

Meaning. (World; witnesses) A course keeps widening where nothing admitted is lost and something keeps being opened (S-21). Take a course keeping widening by name on the actual levels, with what every formation supported from some step on requires held within a finite family of conditions `cs`. The first theorem: the family's stock passes every bound `K` at some step after any `M`. Under the theory's first law, the family without repetition, at arbitrarily late steps the rest of the world gives the family net or the family is not closed (`Web.KeepsWidening.from_rest_or_open`), and a finite balanced world under the theory's first law has no such course (`Flow.whole_not_keepsWidening`). Under upkeep, each supported formation taking at least a unit a step, the course has ever more takers (`Web.KeepsWidening.more`); with the family balanced on from a step, its replenishment, and with bounded giving per entry its renewers, are bounded from no step on, entries counted with repetition (`Web.KeepsWidening.renewers_unbounded`). What carries the openings is fixed by a uniform law of support passing on: "every gain from step `K` on given, up to double negation, by a supported formation that gained support within the `K` steps before"

(ch. 7). The second theorem: under balance, only active formations giving and activity needing support, with requirements never loosening (a local hypothesis, not one of the World's premises), nothing keeps widening along any continuation (a relation saying which formation continues which across a step) containing passing of support (a supported formation giving to a condition another requires, the condition rising and the other gaining support across the step), which also excludes keeping widening alongside the generation of the supported pattern on both readings used, by name and by passing of support; where a list holding every condition any formation requires is dissipated on from a step, nothing keeps widening along any continuation containing carrying on of the supported (each supported formation continuing itself), by name and within kinds included (`Flow.not_keepsWidening_of_dissipated_requirements`). Witnesses show the premises holding together and each needed.

What keeping opening needs instead (the second theorem's premises). Along any continuation containing carrying on of the supported, what is supported descends, up to double negation, from what was supported at every earlier step or from a *beginning* between, a passing of support into a formation nothing supported continues into along it (`Flow.descends_or_begun`). So keeping opening needs a beginning after every step, and by name a passing of support after every step (`Flow.begins_late_of_keepsOpening`, `Flow.passes_late_of_keepsOpening_byName`); not conversely (`KindRelay.Runs.beginnings_without_opening`). Within a kind, a beginning is a passing that begins a line of that kind while the whole lineage goes on (`Flow.BeginsAlong.carries_kind_iff`). On a witness of six formations of two kinds, closed and conserving (`KindRelay.conserving`), the kind chosen so that the passings across kinds begin, the levels keep widening within kinds while what they can form by name only returns (`KindRelay.keeps_widening`, `KindRelay.not_keepsWidening_byName`, `KindRelay.returns`); a fed form is dissipative, its loss paid by fuel no formation requires (`KindRelay.Fed.dissipative`, `KindRelay.Fed.requirements_not_dissipated`). The first theorem's bar on a finite, closed, balanced world is a law by name.

How far seen, and use. Theorems. "The first two laws restate open-systems thermodynamics and maintenance scaling, the others are near-definitional, and none compares two courses" (ch. 7). The method chapter: "a finite family whose kept widening goes on is fed from beyond, and a finite, closed, balanced world has none, as open-ended evolution measures unbounded evolution against an isolated system's recurrence (Bedau and others 2000; Adams and others 2017)." (ch. 8) And "Renewers multiplying under upkeep is maintenance scaling, as in break-even investment (Solow 1956) and the maintenance coefficient (Pirt 1965), which semi-endogenous growth (Jones 1995) leaves out as stated" (ch. 8). Of fed from beyond at the origin of life: "autocatalytic sets are food-generated (Steel 2000; Hordijk and Steel 2004)" (ch. 8).

Known weaknesses.

- "Passing every bound uses natural-number amounts, a finite family having finitely many configurations within a bound; in a dense order needs approaching a limit could support ever more formations at a bounded stock (argued, not proved in Lean)." (ch. 7)
- "The TAP equation and assembly theory leave conservation out as stated, and their own variants and remarks supply it (Cortês and others; Sharma and others 2023), so the law is no third part against them." (ch. 8)
- "None of the laws compares two courses, and none is a result no neighbouring field states." (ch. 8)

- [this dossier, AI] The first theorem reads keeping widening by name: a course keeps opening where, after every step, it admits a formation under a name not admitted at that step. Under the second theorem's premises nothing keeps opening along passing of support (`Flow.not_keepsOpening_along_passings`), so what the first theorem counts as new is new by name, not by descent. Measures of open-endedness count new kinds or lineages. The laws state a beginning, and a witness within kinds given in advance; no law here counts new kinds.

Questions.

1. Is a finite family's kept widening by name being fed from beyond the open-systems condition that open-ended evolution research already states, or does the count of configurations add anything?
2. Is the multiplication of renewers under upkeep a known form of maintenance scaling in growth theory or microbial physiology?
3. Under the second theorem's premises nothing keeps widening along passing of support, and keeping opening needs a beginning after every step. Is that a fair formal reading of "every cell from a cell", and what notion of novelty along descent would a comparison of two courses need?

S-23 The root's laws as the mathematics of simulation and lumping

Meaning. (root) For a relation `R` and a class `E`, *constancy* says that whatever one member becomes, a member alike to it becomes something alike; the *core* is the greatest symmetric relation inside `E` constant under `R` (`«根»`). "So the core is the greatest matching relation within the class, and it always exists." (ch. 1) "For a class this is bisimulation equivalence, with the lumpability of Markov chains and computational closure as analogues not proved here." (ch. 1) The record's credit: "the closure law is the relational analogue of lumpability (Kemeny and Snell 1960) and of computational closure (Rosas et al. 2024), a correspondence not proved here." (a result record) Beginning, abiding and ending are read off the two-way core (`«位根»`). Since 31 October 2026 the core is also set against going alike, the relational form of a lumping: where the step's successor is fixed, the core of a class is a class inside it that the step goes alike across, and with every occurrence having a successor it is the coarsest such class (`«定后遍后则核为其内最粗同行之类»`); off a fixed successor it need not go alike (`«后不定则核可不同行»`).

How far seen, and use. Theorems (root). A search's second round: "Every root law holds in every model, so it forbids no observation." (a study record) "So the core's laws are bisimulation and partition refinement" (a study record). The logic dossier treats the core and closure (L-06, L-07), crediting Park (1981), Milner (1989) and Paige and Tarjan (1987).

Known weaknesses.

- "A concurrency theorist given the identification above would foresee each one." (a study record)
- [this dossier, AI] The laws are relational. Lumpability of Markov chains (Kemeny and Snell 1960) needs a probabilistic notion, which the chapter calls an analogue, not proved; its exact counterpart is probabilistic bisimulation, as the logic dossier says: "Lumpability (Kemeny and Snell 1960; as probabilistic bisimulation, Larsen and Skou 1991) asks equal transition-probability totals from a block's members into each block, which nothing here expresses" (LOGIC).

Questions.

1. Is the core a useful notion of the level at which a pattern lasts, compared with lumpability or the causal states of computational mechanics (Crutchfield and Young 1989)?
2. Would a probabilistic version be needed before the root says anything about stochastic systems?

Response form

For each item give one verdict: **accept** (the item is correctly described, credited and argued at its strength), **accept with changes** (say which), **reject** (say why), or **cannot judge**. Comments may answer the numbered questions by number.

Item	Verdict	Comment
S-01 The verdict: restated or near-definitional		
S-02 Third parts: farther than a few frameworks, not than the sciences together		
S-03 Strength as content: the theses and the unifications		
S-04 Balance and the least loss, read from thermodynamics		
S-05 Recurrence, merges and Landauer's count		
S-06 Rest needs a least loss; continuing change is fed from beyond		
S-07 The census and the classical laws in counts		
S-08 What bounds offspring numbers: Price and Fisher joined to the flow		
S-09 The pace of selection within what flows in		
S-10 Loops and regulation		
S-11 Joint excess under a gated renewal		
S-12 The margin of a regime		
S-13 The optimum laws for a renewable resource		
S-14 T1 and its tests		
S-15 T2 and its tests		
S-16 T6: a break seen coming		
S-17 The free-energy principle as a case		
S-18 Conditioning by restriction and the exit from a margin		
S-19 Orders with a set point		
S-20 Selection through couplings across fields		
S-21 Eventual widening		

Item	Verdict	Comment
S-22 What keeping widening needs		
S-23 The root's laws as the mathematics of simulation and lumping		

General. What would a scientist in your field add or object to, beyond the items? Is any known result restated without credit? Which framework of your field would a containment have to forbid something of before you counted it as reaching farther?

Returning the form. Fill in the table (a copy of this file, a printout or a list by item number all serve), add any longer comments, and return it to whoever sent you the packet. Say whether your comments may be quoted, and under what description of your field.

How this dossier was made

Everything in the theory was made with AI: the Lean proofs, the chapters, the readings, the tests and this dossier. No independent human review of it is recorded; the person who directs the work has read its important parts, not every word.

- A script copied the Lean statements from the pure core (`lean_statements.py`), and a second compared every block with its source, whitespace ignored (the kit); the periphery's laws (S-04, S-16, S-17) are named, not copied. The packet's script checks every quotation against the file it names and every backticked Lean name against the core and the periphery's `lean/physics` and `lean/warning` (`check_dossiers.py`).
- The prose is AI-written and may be wrong. An independent AI reader reviewed it in two passes, every finding taken, and read its revision of 2 November 2026 (the kit). On 3 November 2026 the whole was read against what had changed since 27 October, and another independent AI reader checked the change and the other flagged items against the chapters and the Lean; its fourteen findings were checked against the sources and taken (the kit). At the freeze the same day it joined the first edition: it moved into the review folder, its links re-pointed, and its hour, whose seven items ran past an hour at 100 words a minute, became S-02 and the reader's field's items, a path that for ecology, the longest field, ran to 65 minutes at that rate, past the hour by the lines on T2 that follow; S-15 and the front's T2 row took T2's sentence on the life cycle, landed before the freeze, on what UE31's frozen test expects in the flour beetle, and the test's run, landed just before the freeze, which decided nothing: S-15's bullet on the beetle now ends with the result in 02's words, in place of the bullet that landing added. An independent AI reader then read the whole freeze, this joining included; its findings on the joining were checked against the sources and taken: the cover's row and the covering letter now ask what this dossier asks, the letter's table names this response form, the repository's map and chapter 8 now list this dossier, and the kit no longer says it is not offered. Another AI reader, read-only, checked S-15's and the front's new lines on T2 against 02, 08 and UE31's design; its findings were checked against them and taken: the clause expected to fail is named, not hidden by an omission, what would remain and why a culture off the *Refuted by* list counts are quoted, and 08 §9's reading is given; its two findings on the homes went to T2's owner, who took both before landing: the sign's condition, and MQ6's constant dilution outside the settings T2's *Refuted by* names rather than outside those that

could refute it. S-15 follows both, and its own reading of that run now rests on what the run measured. A second AI reader, read-only, read the lines again after T2's owner revised the homes, and then the fixes; its findings were checked against the homes and taken: the beetle bullet gives the design's sentence on the field's state before 02's, so that 02's "Their" follows the three numbers, and quotes the design's numbers and condition on the sign, its terms glossed; the front's row follows 08's, with the lag where the life cycle puts it; and the dossier's reading of MQ6 names the tension it found between T2's second forbidden item and the law's regime's-step end, asked as question 4. Its two points on the homes, that tension and the sign's dropped condition, went to T2's owner; the condition is in 02 since UE31's run landed. Later that day (HP) the ecology path was brought within the hour, cutting only what the dossier says more fully elsewhere, in S-14, S-15, S-16 and the front's T2 row, and leaving S-02 to the other fields' paths, the front's verdict taking S-02's caveat on the verdicts against the literature in S-02's words, shortened. An independent AI reader, read-only, then read the change; its findings were checked against the sources and taken, among them which cultures UE31's run found too short, now on the front's row, and an hour sentence that is an instruction for every field (what was cut, and why). That evening (TS) the front's table gave each thesis its standing in the words of chapter 8's table of theses, and the line after it T3's. Another AI reader, read-only, checked these against chapter 8 and the theses' homes, and then the fixes. Its findings were checked against them. Those taken include T3's standing across domains, from chapter 8's table of unifications, and two seabird tests sharing rows. Two on T1, whose home then gave its standing only in the model, went to the coordinator for that home and chapter 8's row. The rest were left, with their reasons (what changed, and why). Once T1S had given T1 its standing there, the front's T1 row, S-03 and S-14 followed it (TS2). The same AI reader checked them against the home and chapter 8's row, and then the fixes. Its findings were checked against them and taken, among them qualifiers on T1's passes that the front had dropped and the time of ecology's path, which ran past the hour again; one on the home's own wording went to its owner (what followed). On 4 November 2026 (TS4) the front's T1 row bounded the point's exactness to a top the step keeps, as its home does: T1's home's owner had shown, with giving that overcompensates, levels above such a top that end though the point does not. The words are that owner's, checked against the home and the Lean by another AI session (the point's top). The same day (SF) the front was cut toward its ten minutes only where it or the fuller part already says the same; every standing, failure, qualifier and premise stays, with what is not claimed, and the section states the time it takes, the excess recorded. Another AI reader, read-only, checked the cut against the fuller part, and then the fixes; its findings were taken (the cut). On 4 November 2026 (T2S) the front's T2 row and S-15 took T2's statement as its home restated it from UE32's proposals: the second forbidden item set against the three numbers, read at the grain fixed before the data and lifted only in a series a test shows, by a bound, to sit at the regime's-step end, the lag's place named before the data, and question 4 asked of that answer. An independent AI reader, read-only, read the change against the proposals and the study, and then the fixes; its findings were checked against the sources and taken or argued (the reading). On 4 November 2026 (TS3), once T1P had given T1's exact point its premises in the home and chapter 8's row, the front's T1 row and S-12 followed, kept for the batched pass and taken in edition 5 on 7 November 2026. Another AI reader, read-only, checked them against the home, chapter 8's row and the Lean, and then the fixes; its findings were checked against them and taken, among them what the point's exactness and the rises' superadditivity are proved under, and one on the home's theorem on the rises went to its owner (what followed). The verdicts against prior work quoted here come from AI reviews against the literature an AI knew; they are what a reader is asked to check.

- Read again for edition 2 of the packet, on 4 November 2026, against what the chapters had changed since edition 1's frozen commit. Since that commit, other passes had edited the dossier on 3 and 4 November: its quotations on the proxy problem, the outrun record and EM1, real amounts, Hamilton's rule and the breeding value, the theses' standings and T1's exact point, T2's statement, the free-energy principle's place in 08's verdict and UE32's run, and the front toward its ten minutes. This pass added S-01's pointer to 08's last column, its drift and evolution rows as 08 gives them, S-07's group laws and count-weighted chain, S-12's rises law, S-13's coupled drawers and the dated notes in S-06 and S-07. An independent AI reader with no part in the writing read the changes and the flagged items left unchanged, in three rounds; each finding was checked against the sources and taken, but for three already listed for one batched pass after readers report, one passed to 08's owner, one optional addition that would lengthen ecology's hour, and the stated times, set by the measure. After its last round the front's T2 row took UE32's result in 08's words, read against 08 by the dossier's holder alone.
- Read again for edition 4, on 6 November 2026, after edition 3, against what had changed since edition 3's frozen commit. UE31b's run failed T2's former wording in the flour beetle's adult-loss cultures, so the front, S-15 and its second question give T2 as chapters 2 and 8 now word it, and what remains and why it is no pass. S-11 gives chapter 5's sentences on what is proved at and after the step the excess reaches, among them its reading of chapter 4's lineage law with no gate; this dossier's assessment of the gate says what depensation bounds after that step, and that at a need of one, counted in the least that can renew, it bounds nothing, a flaw edition 3 shared, as chapter 2's sentence on depensation did until corrected for this edition. T1N and T1R each ran once and refused before reading a row, so S-14, the front's T1 row and S-14's first question say so. An independent AI reader with no part in the writing read these changes in three rounds, stopping at the first with no high or medium finding; each finding was checked against the sources, and those that stood were taken, one narrowed.
- Read again for edition 5, on 7 November 2026, after edition 4, against what had changed since edition 4's frozen commit, with the board's batched precision pass. The front's T1 row and S-12 give T1's exact point with its premises, as chapter 2's T1 and chapter 8's row word them since T1P (TS3, kept for that pass), and S-12's rises the regime record's correspondence without "harm", with what MQ1 measured: harms read by removal, not the point's rise. The front's T2 row and S-15 give UE32's frozen criterion and, post hoc, its figure within the field's bound, in 08's words. The verdict line, S-02 and S-17 give 08 §8's verdict on the free-energy principle as 08 and the principle's page now word it: farther than the principle alone in what it forbids, less where the theory states nothing yet, and that a theorem a specialist would not foresee "could", not "would", change the verdicts. An independent AI reader with no part in the writing read these changes in three rounds, stopping at the first with no high or medium finding; each finding was checked against the sources and taken, among them two that corrected chapter 8 and the principle's page and this line, which its first round asked for, but one on a dated note outside the dossier, left as written.
- Read again for edition 6, on 10 November 2026, AI-written, after edition 5, against what had changed since edition 5's frozen commit, on the text the AG review left and with the board's batched precision pass. S-12 gives where a run turns (02, landed that day): the two levels a turning run names, the band laws, and the period-three witness outside their premise. The front's T3 row gives the simulation before T3 was worded, its T2 row MQ5's verdicts on the state and the lag apart from the two on the draw's response, and the free-energy principle's five statements as two of them weakly; S-01's credit of 08's additions and S-21's second-order direction say what 08 and 07 now say. With the pass, the

verdict line and S-01's quotation of 08 §8 name the premise ledger as new only as an artefact, the quotation of 08's table Zhuang and Hadfield-Menell's Theorem 1 at optima rather than limits, and S-14's quotation of T1 the count of renewal cycles each test fixes before it runs. Independent AI readers with no part in the writing read these changes: before the review landed, one for this dossier, whose findings were checked and taken; on the text carried onto the review's, another, who also read the pass's chapters, fronts, data and records, in two rounds, stopping at the first with no high or medium finding; each finding was checked against its source and taken. The front, its hour and the whole are restated as measured, thirteen to seventeen, 65 to 85 and 176 to 230 minutes.

- Under **Known weaknesses**, a bullet quoting a chapter, a record, a study or a test's account reports a recorded weakness. One marked [this dossier, AI] is this dossier's own assessment, recorded nowhere else, for the reader to confirm or reject.
- Works cited outside quotations were checked against a publication record (the kit). Works named inside a quotation are the quoted text's citations.